

Evaluation for Open Ended Generation

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M.Sc. Scientific Computing (8LP)

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MAUVE: Measuring the Gap Between Neural Text and Human Text using Divergence Frontiers

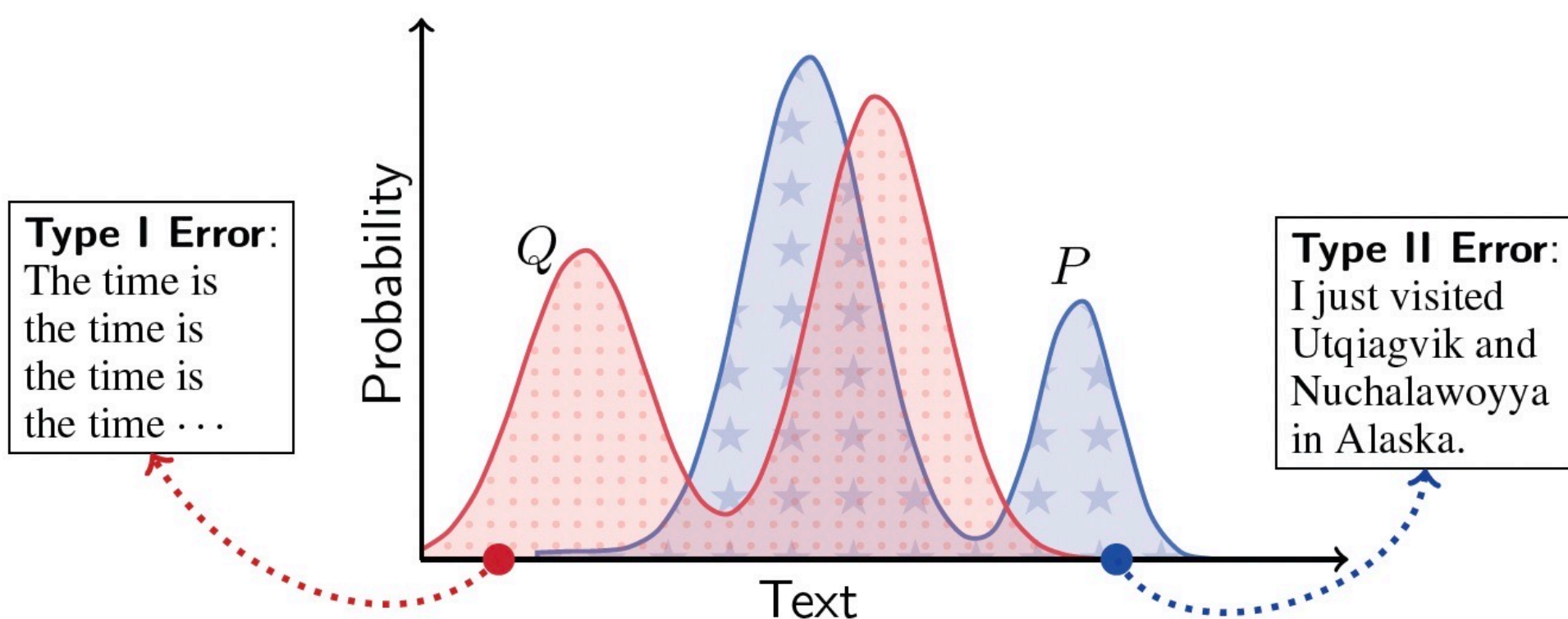
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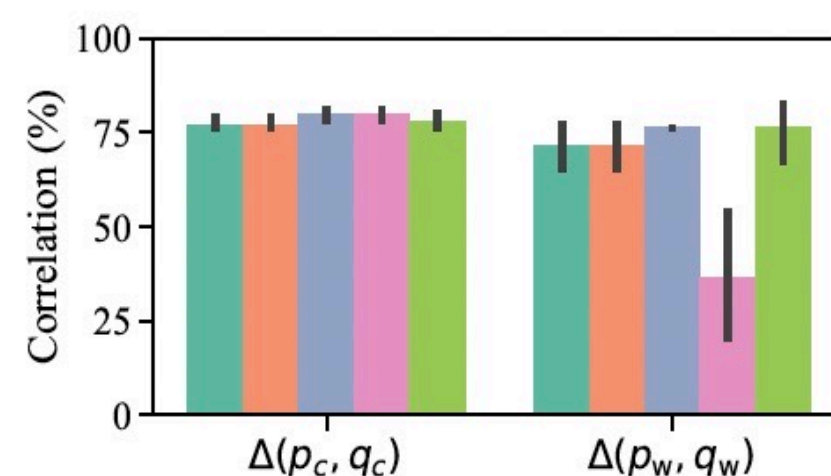


ON THE USEFULNESS OF EMBEDDINGS, CLUSTERS AND STRINGS FOR TEXT GENERATOR EVALUATION

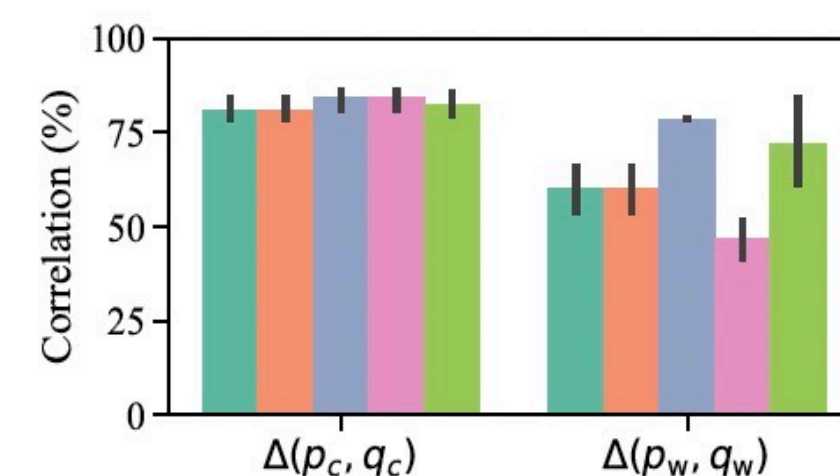
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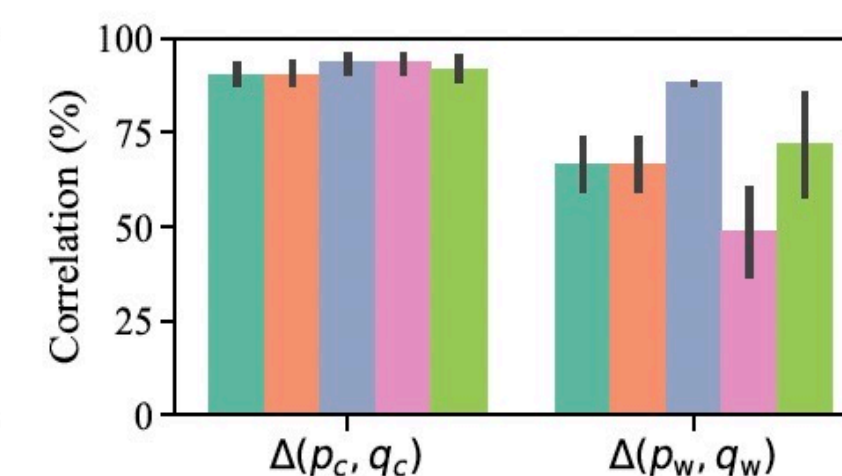
tp472@cam.ac.uk, {clara.meister, ryan.cotterell}@inf.ethz.ch



(a) Interestingness



(b) Sensibility



(c) Human likeness

Text Generation

Large Language Models => Massive Progress in NLP !



GPT



BERT



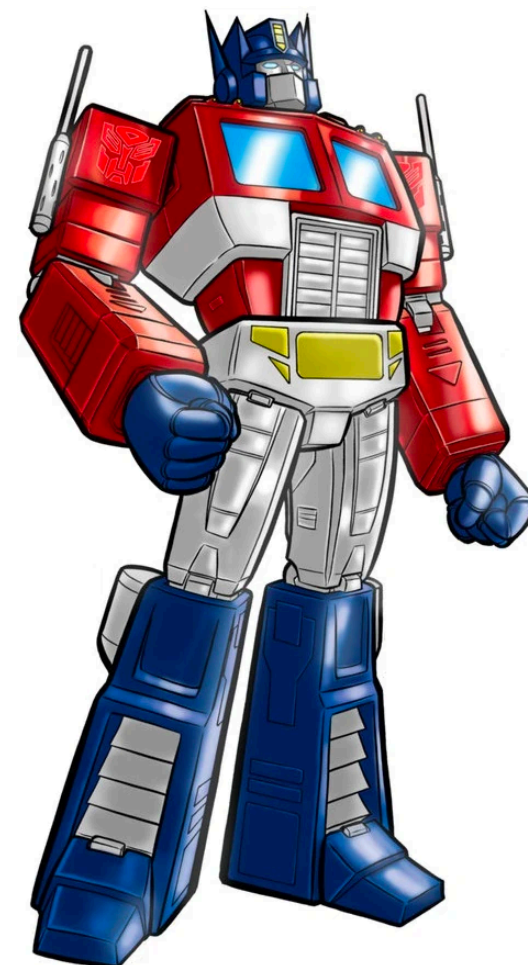
BART



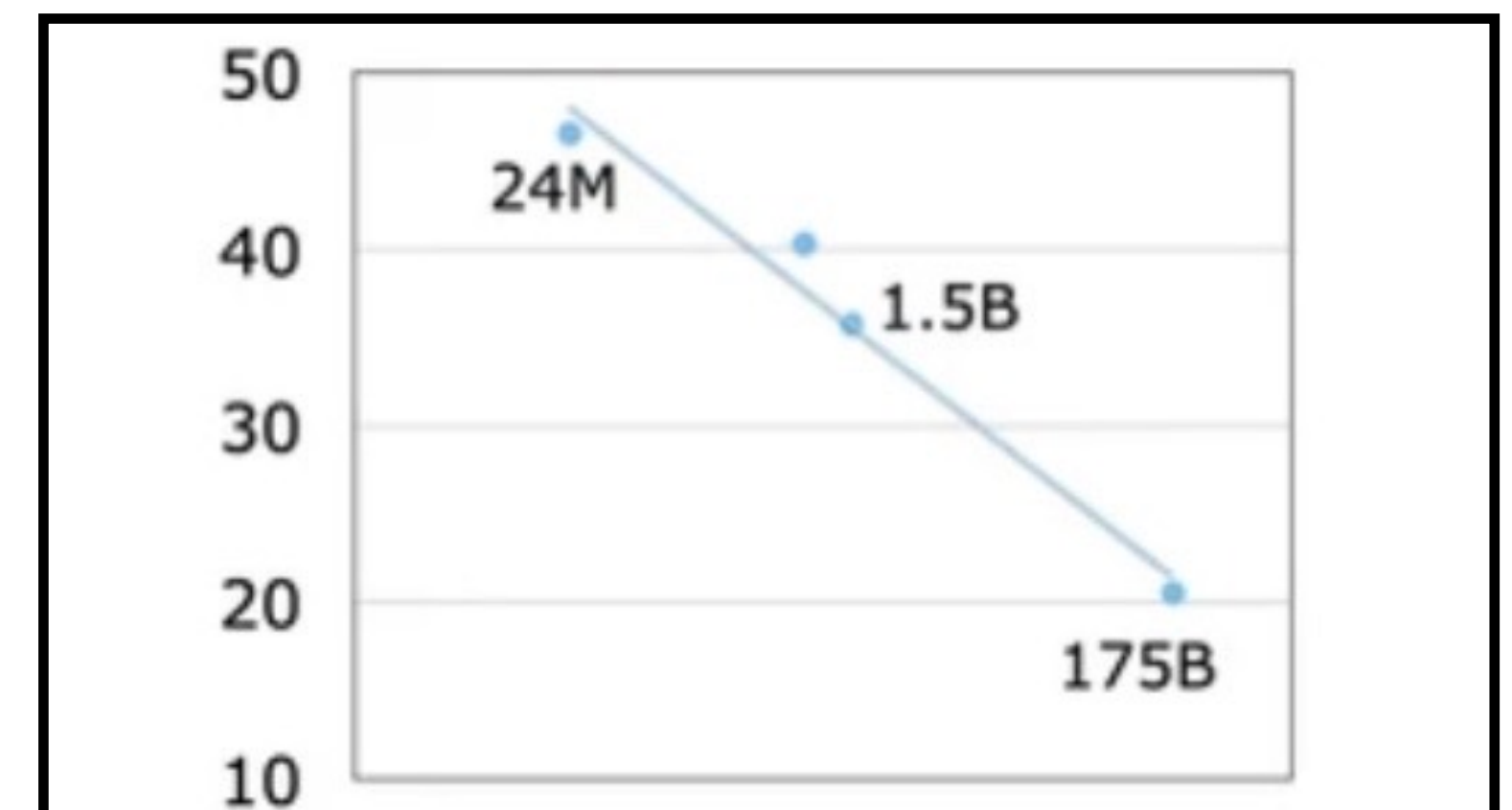
ELMO



ERNIE



Perplexity on PTB



Model size

Open-ended text generation

Large Language Models can write long essays !

 **You**

Be creative! Append after this!

In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

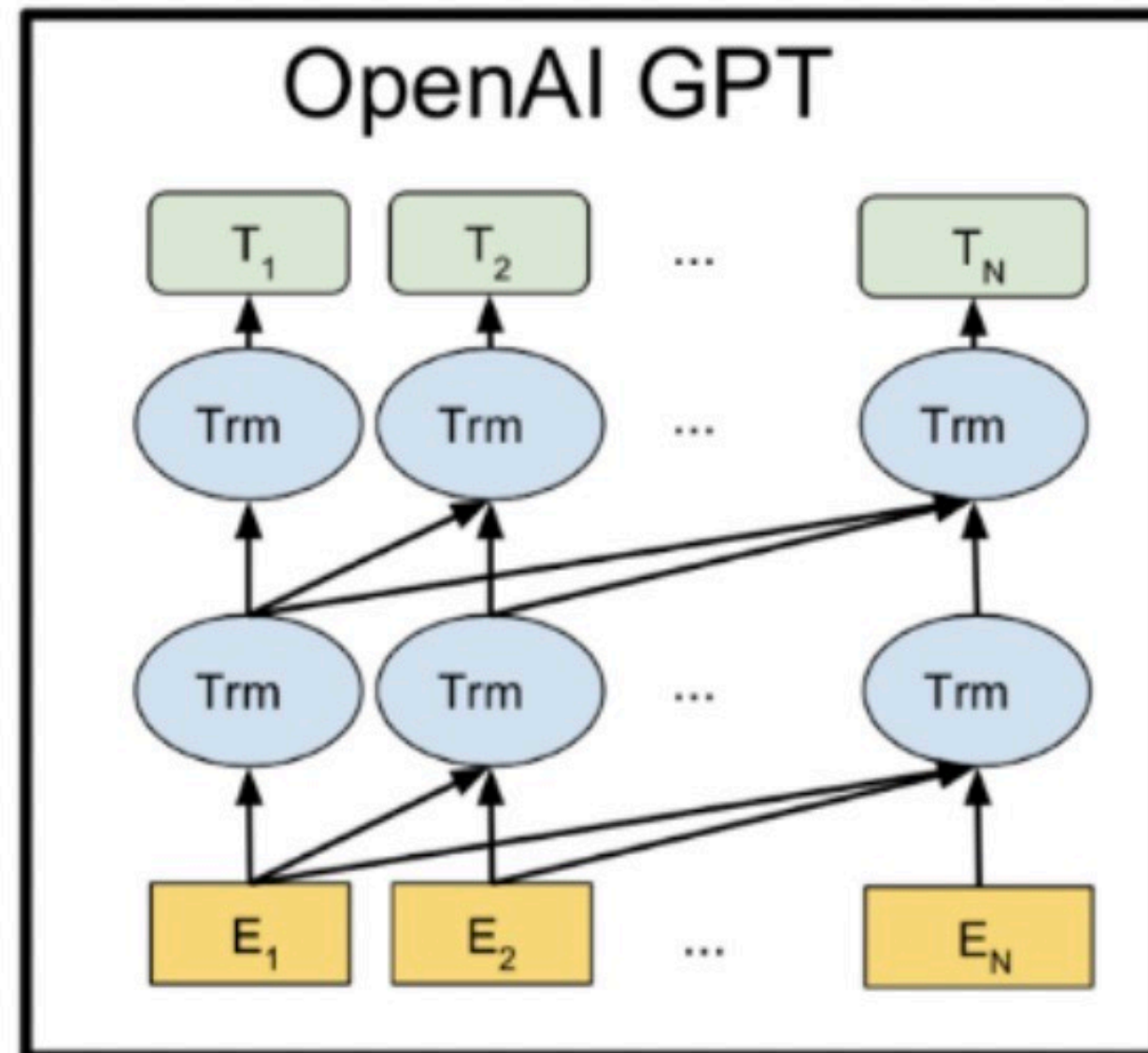
 **ChatGPT**

Not only were these unicorns fluent in English, but they also possessed an astonishing knowledge of human history, literature, and pop culture. As the scientists cautiously approached the majestic creatures, they were greeted with eloquent discussions on Shakespearean sonnets, debates about the ethical implications of time travel, and even references to the latest blockbuster movies.

LLM's can still make mistakes. But how close is to to human text?

Left-Right LM

GPT 3
(Brown et al. 2020)



$$P(\mathbf{w}) = P(w_1) \times \dots P(w_n | w_1 \dots w_{n-1})$$

Auto-regressive

Objective : Standard Language Modelling

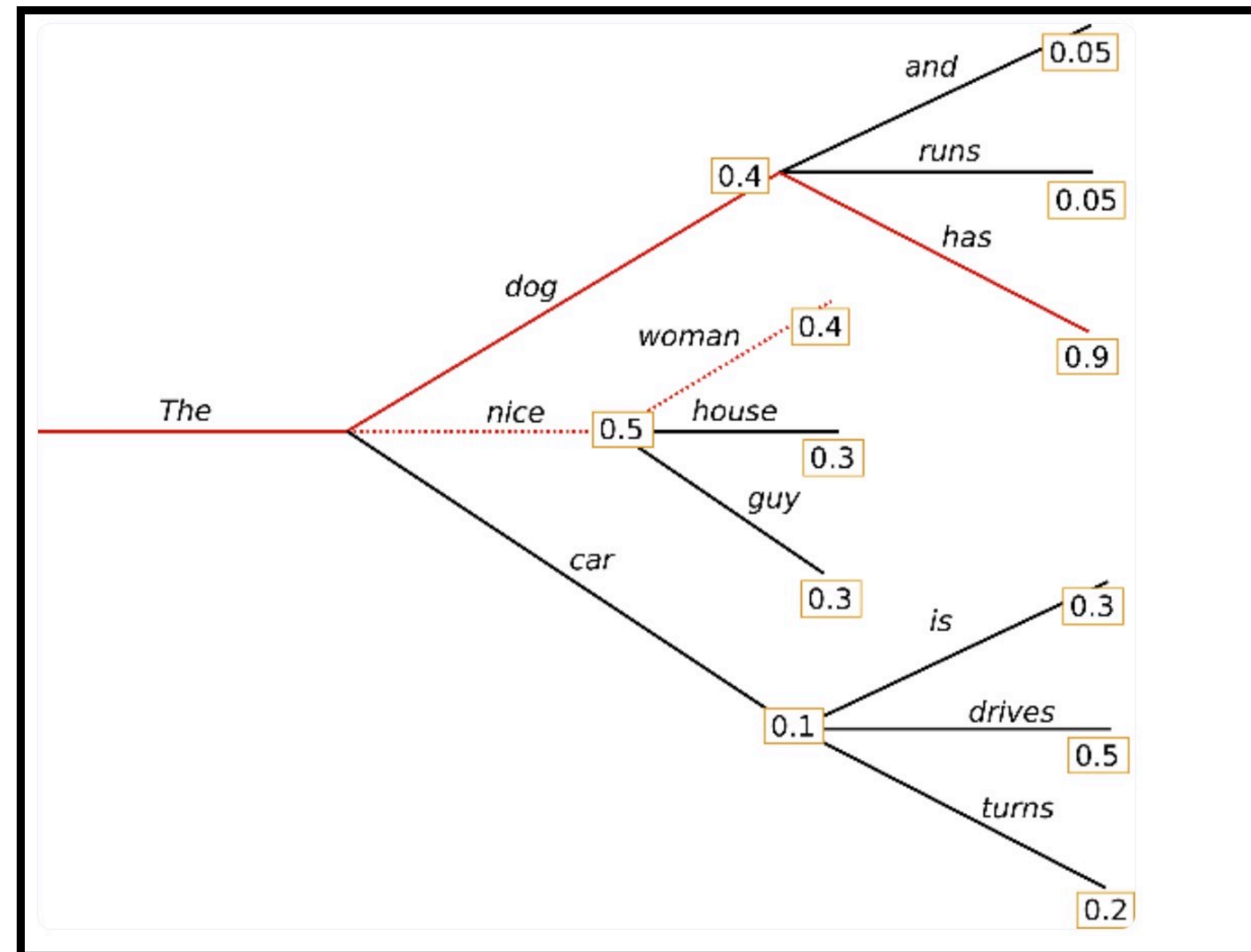
Data : CommonCrawl (1T words)

Maximization based Decoding

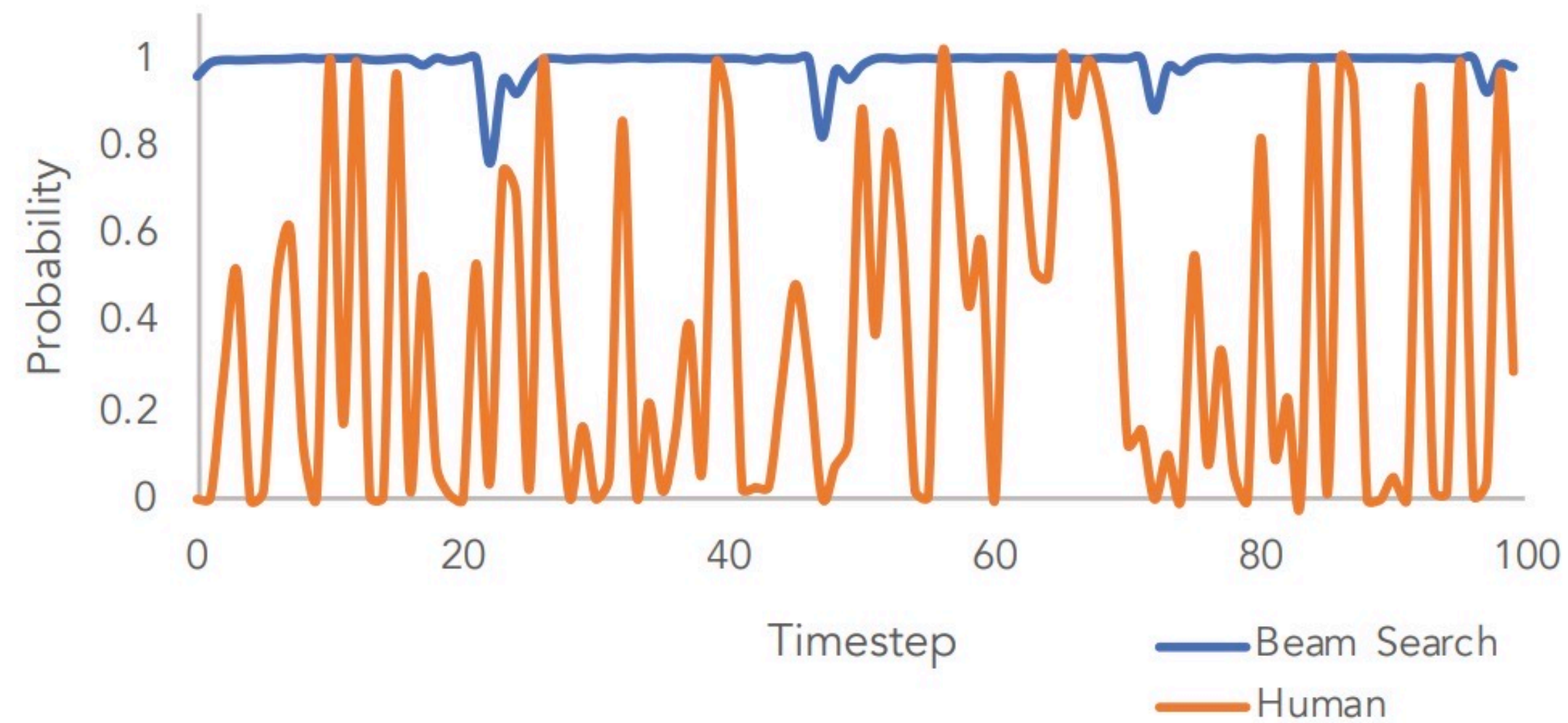
$$\operatorname{argmax}_{\mathbf{w} \in \mathbb{W}} P(\mathbf{w}) = P(w_1) \times \dots \times P(w_n | w_1 \dots w_{n-1})$$

Intractable

Beam Search



Beam search is less surprising !



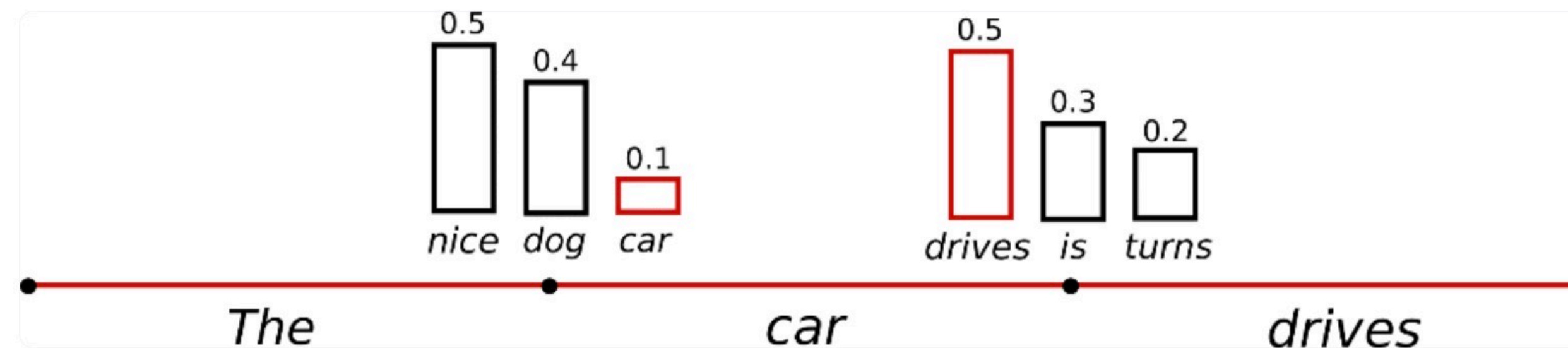
An unprecedented number of mostly young whales have become stranded on the West Australian coast since 2008.

The number of stranded whales has increased by more than 50 per cent in the past year, with the number of stranded whales on the West Australian coast increasing by more than 50 per cent in the past year. The number of whales stranded on the West Australian coast has increased by more than 50 per cent in the past year, with the number of stranded whales on the West Australian coast increasing by more than 50 per cent in the past year.

Sampling !

$$w_i \sim P(w_i | w_1, \dots, w_{i-1})$$

Non Deterministic

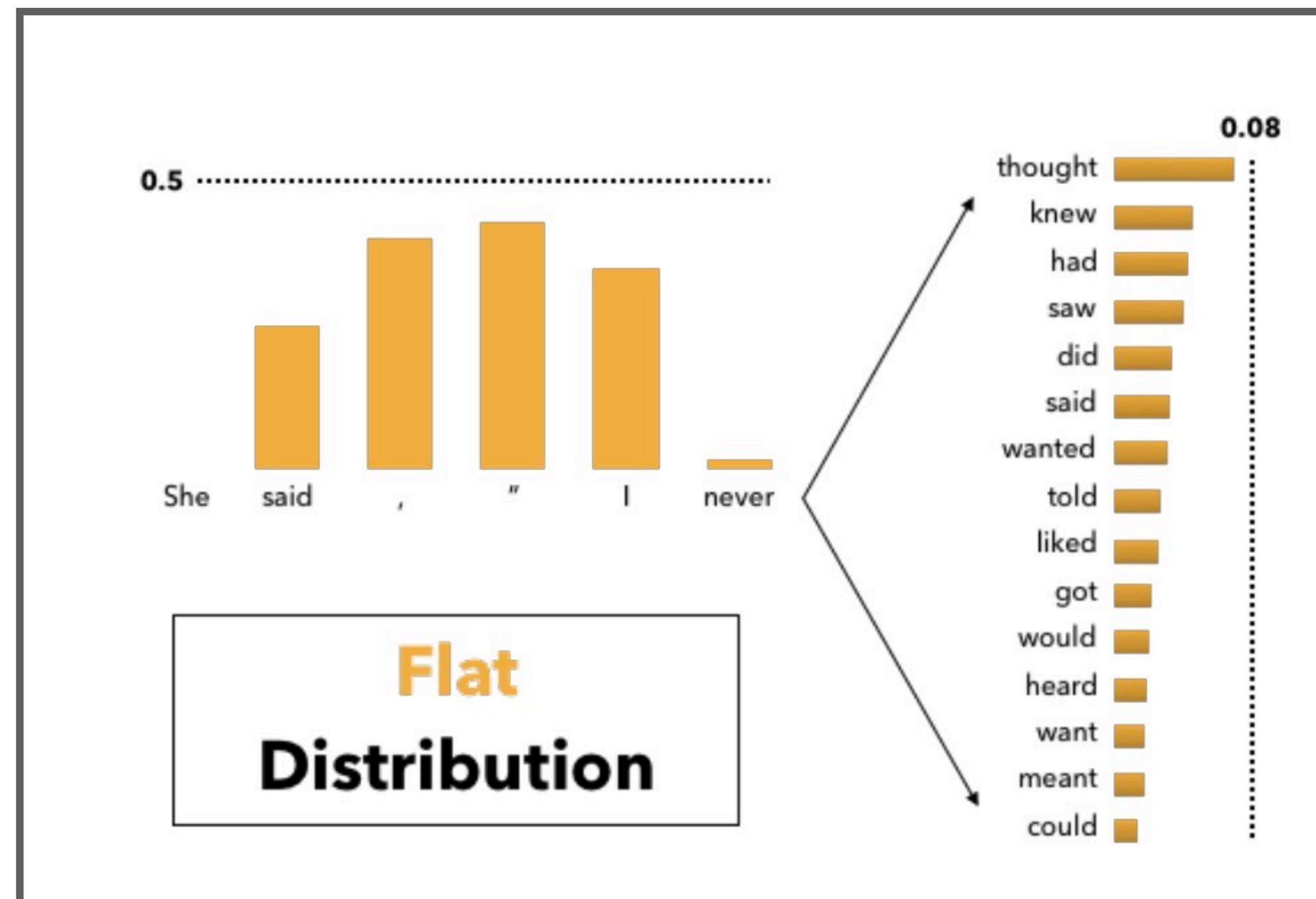


Cross-Entropy has the unfortunate property that distribution with long tails are often modeled Poorly with too much weight given to unlikely event

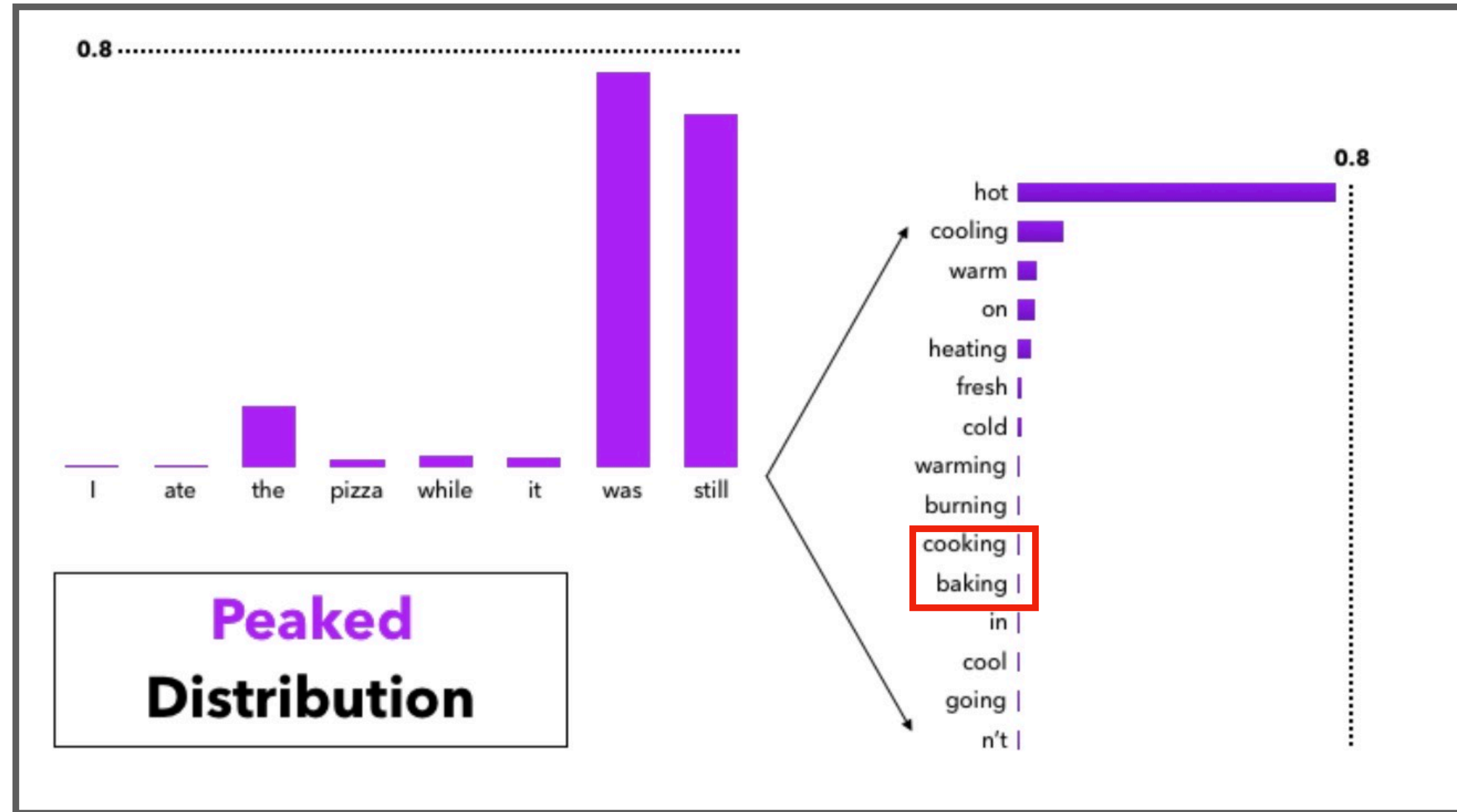
Top K sampling

$$w_i \sim \text{best_k_options} \quad P(w_i | w_1, \dots, w_{i-1})$$

Top K can be too generic !

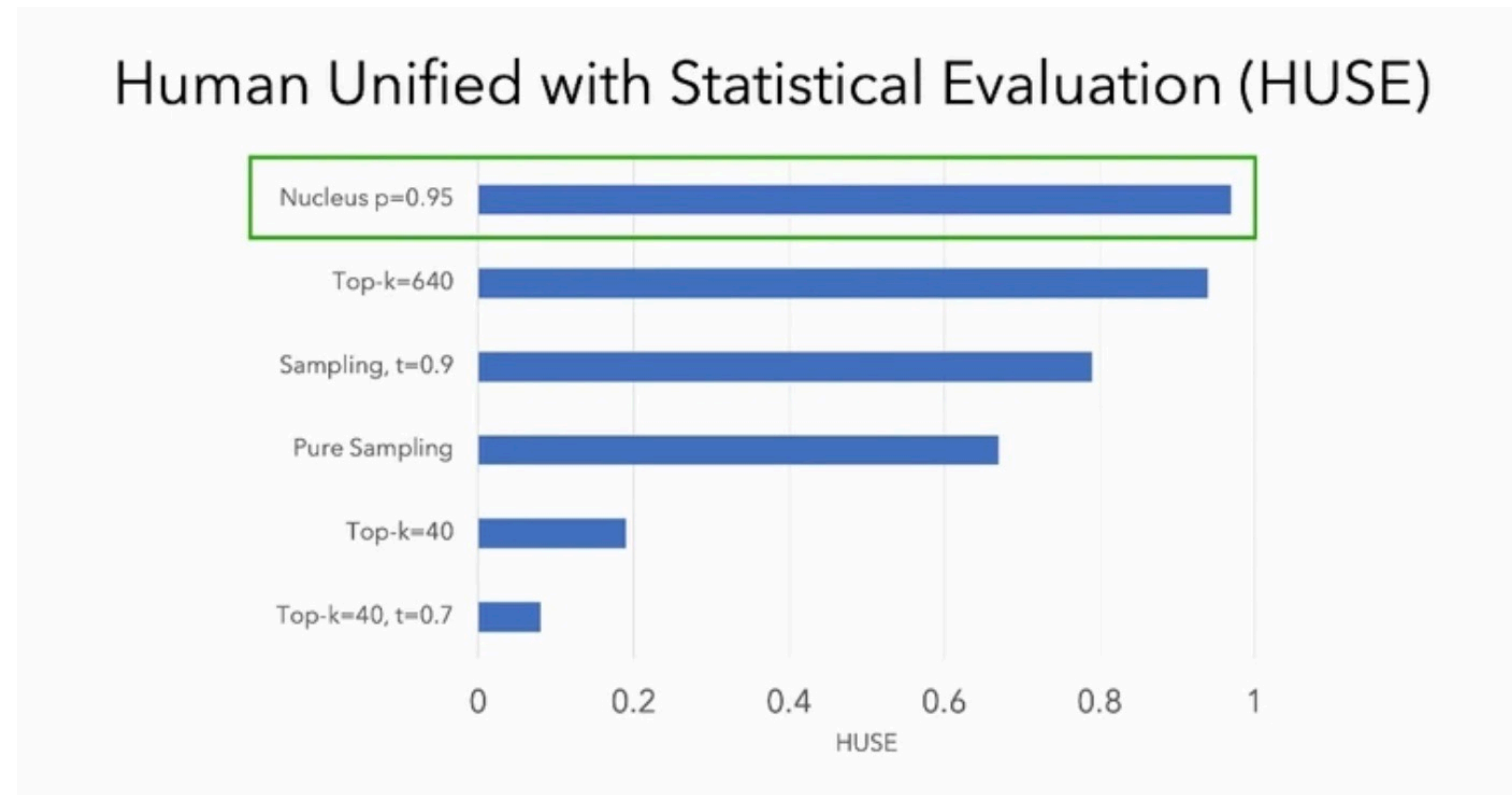


Top **K** can be too random !



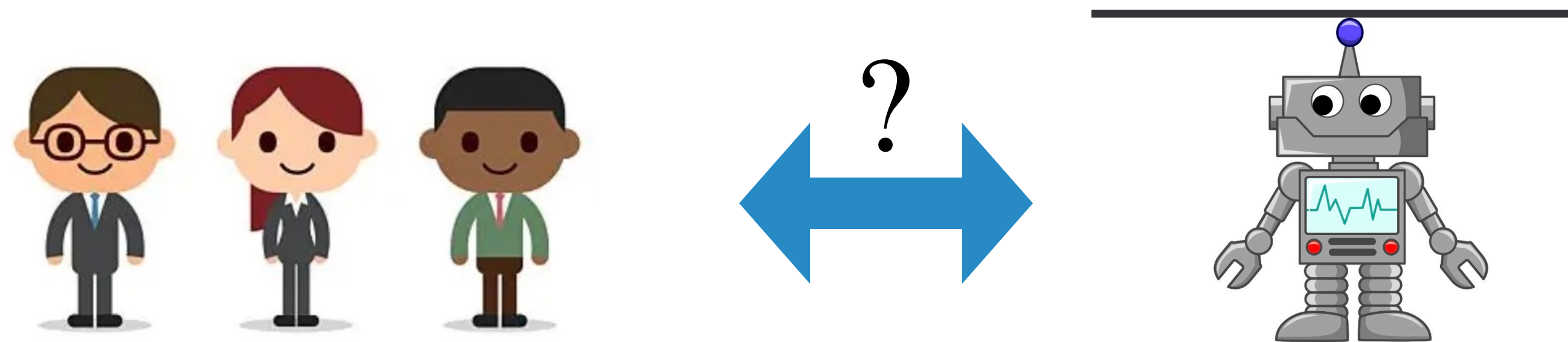
Nucleus (Top-p) Sampling

Top - k sampling where k changes at every time step to cover p probability mass



Machine text vs Human text : How to compare them ?

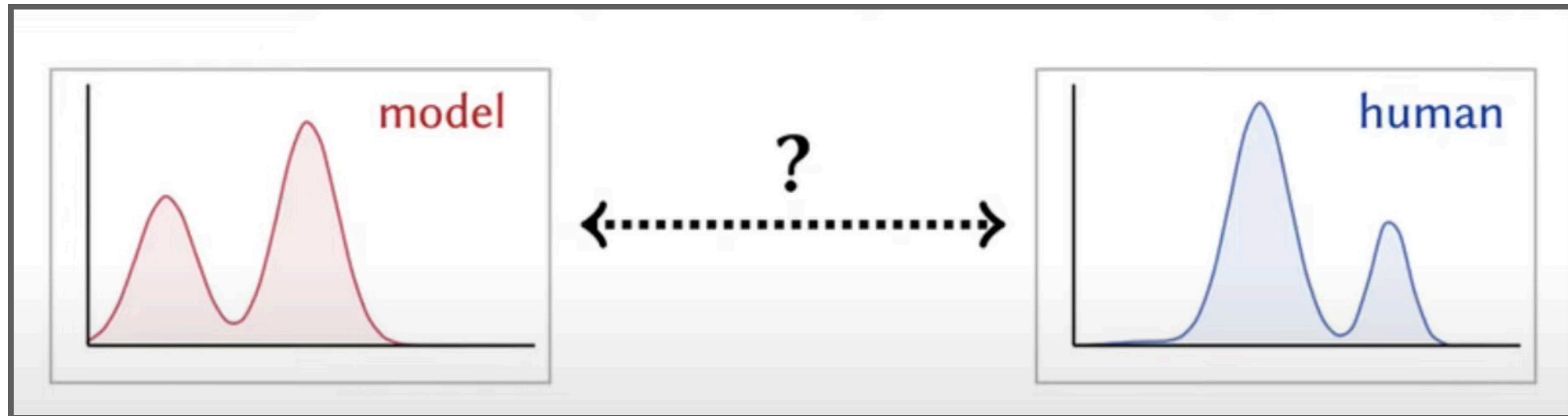
Multiple “correct” answers! Traditional Evaluation based on gold references do not apply



Come up with a statistical diagnostic tool !

Machine text vs Human text : How to compare them ?

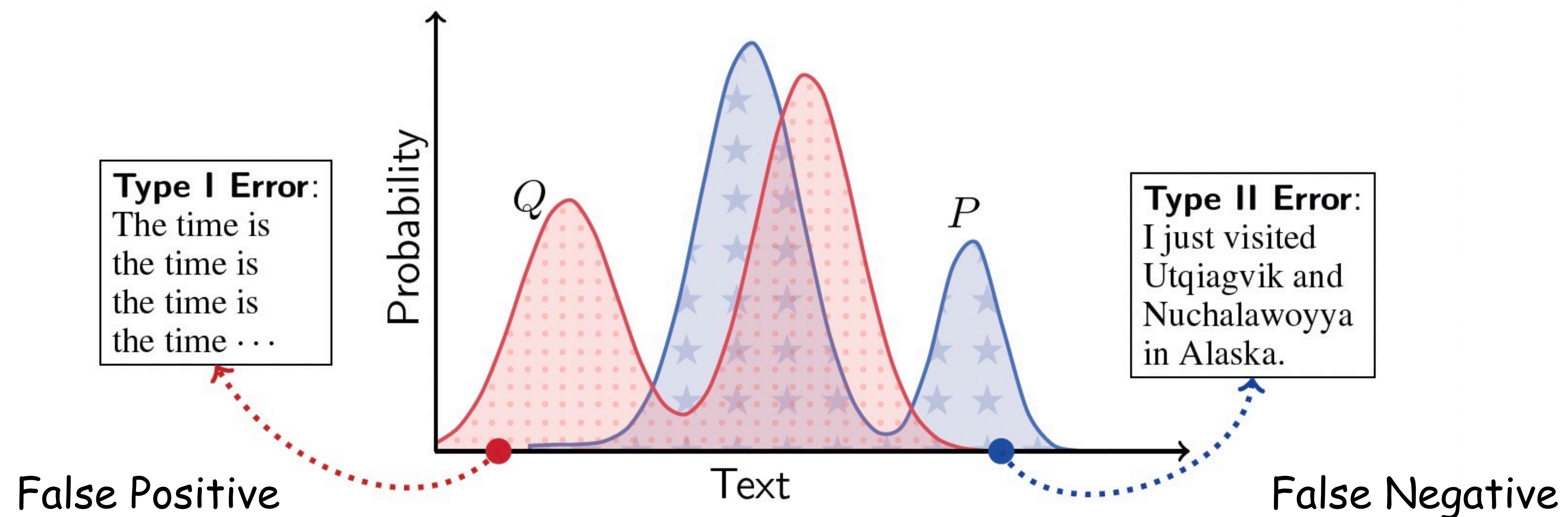
Goal : measure how close two distributions are !



Two types of error in text generation

Q: Machine distribution

P: Human distribution



Q places high mass on text unlikely under P

Q cannot produce text plausible under P

example : degenerate text

example : due to nucleus sampling

We need a “distance” metric !

Forward

$$KL(P | Q) = \mathbb{E}_{x \sim P} \log \left(\frac{P(x)}{Q(x)} \right)$$

Penalizes **P** if there exists text x s.t

P(x) is large but **Q(x)** is small

Type II error

Backward

$$KL(Q | P) = \mathbb{E}_{x \sim Q} \log \left(\frac{Q(x)}{P(x)} \right)$$

Penalizes **Q** if there exists text x s.t

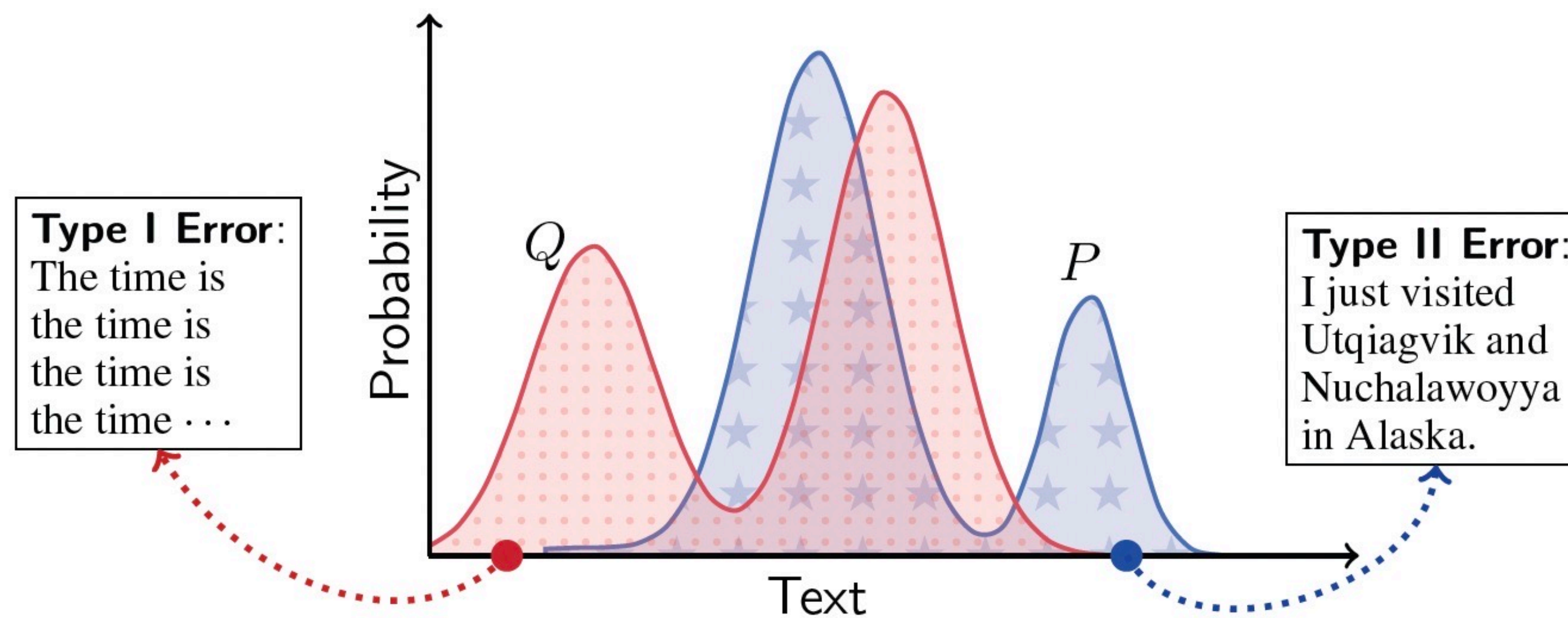
Q(x) is large but **P(x)** is small

Type I error

Two types of error in text generation

Q: Machine distribution

P: Human distribution



$$\text{KL}(\textcolor{red}{Q} \mid \textcolor{blue}{P})$$

$$\text{KL}(\textcolor{blue}{P} \mid \textcolor{red}{Q})$$

Problems !

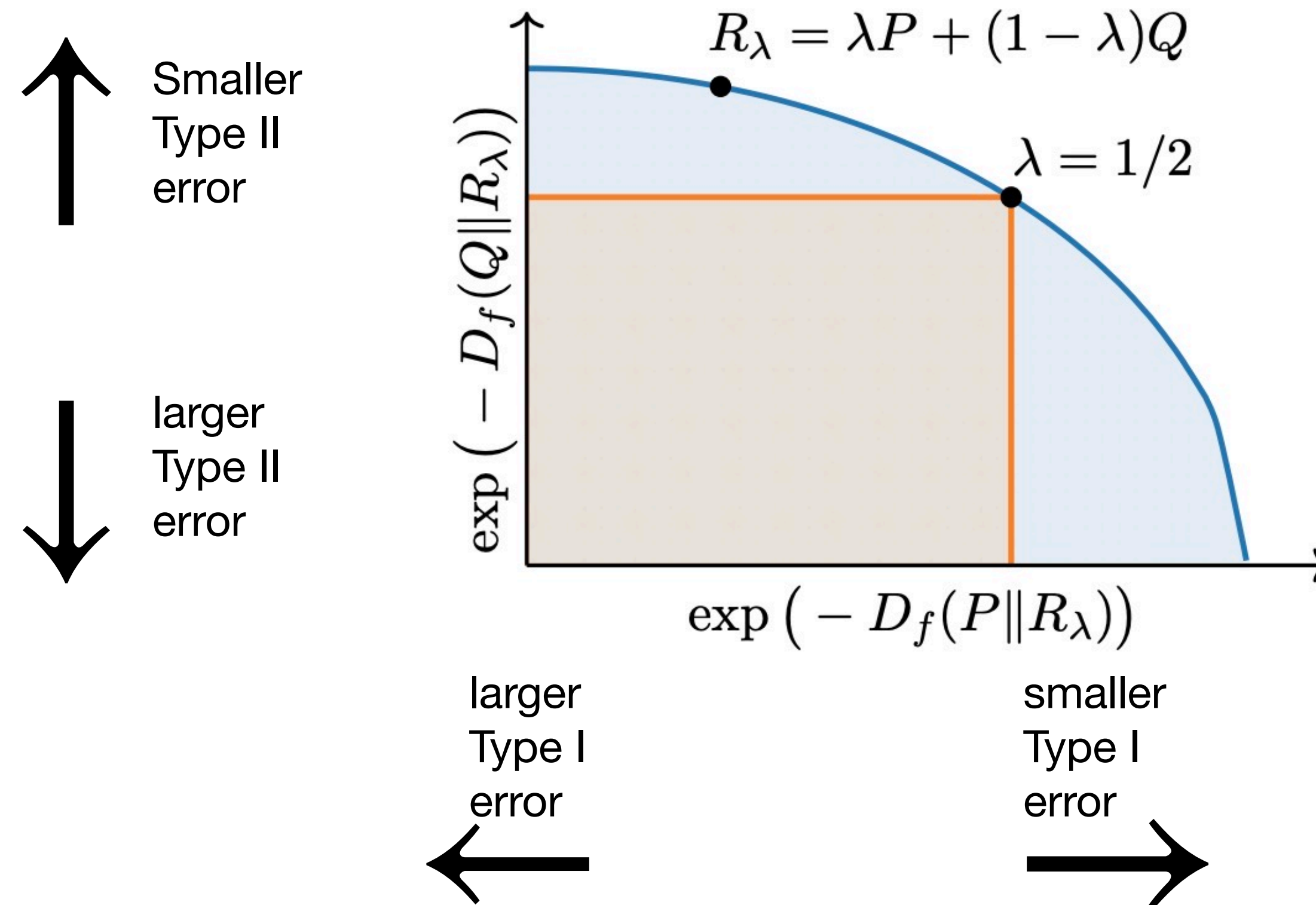
$KL(P|Q)$ or $KL(Q|P)$ can be infinite when supports of **P** and **Q** are not identical



When using sampling methods

Solution : Measure errors softly using mixtures **P** and **Q**

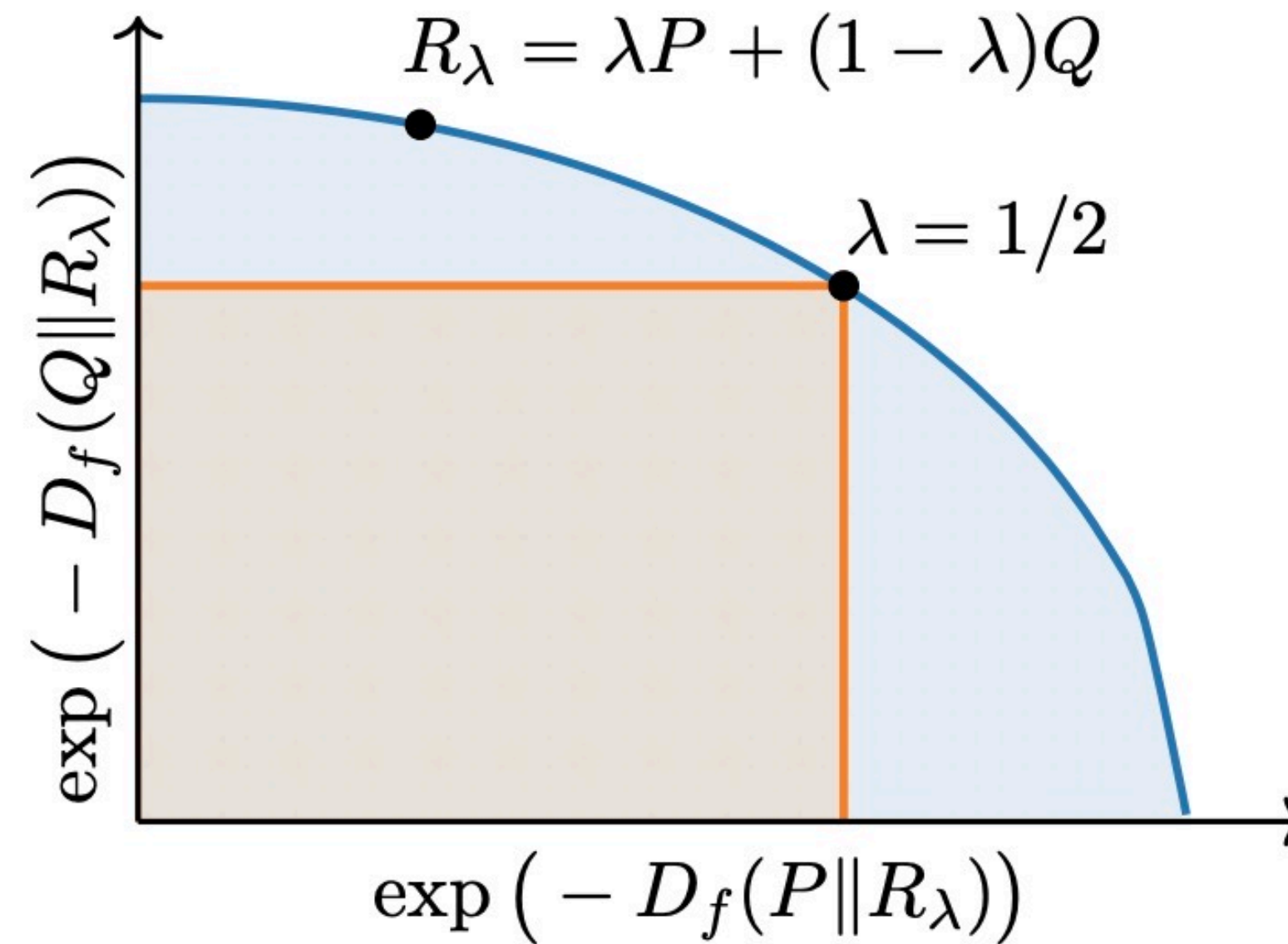
MAUVE : summarizing both errors



Divergence Curve : Varying mixture weights captures trade-off between **Type I** and **Type II** error

MAUVE

Area under the blue curve



1 : **P** and **Q** identical,
0 : **P** and **Q** dissimilar

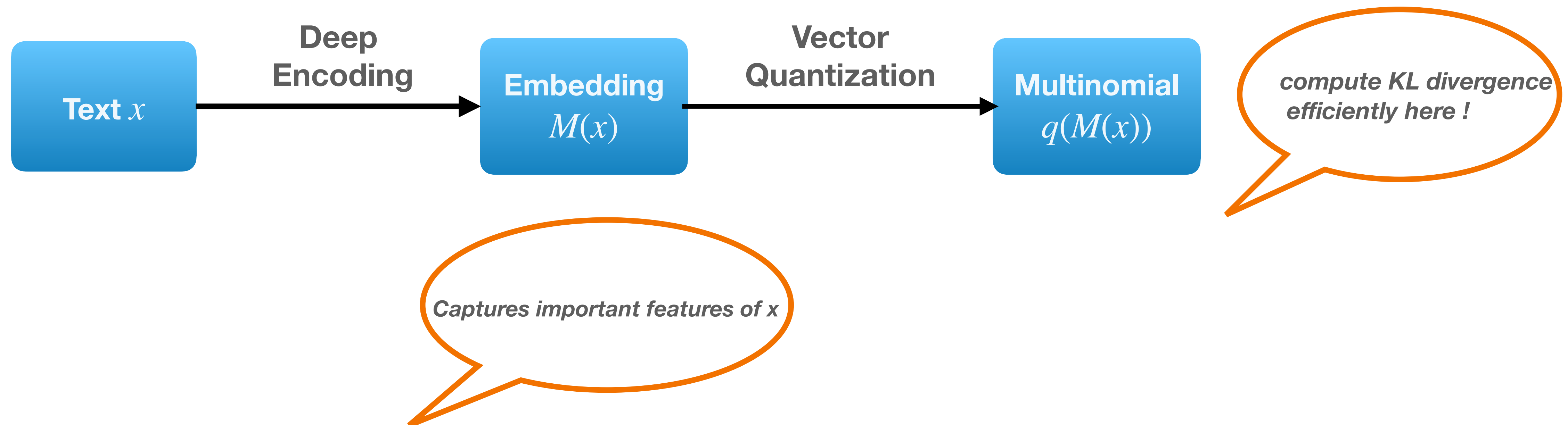
Computation is intractable

$$KL(P \mid R_\lambda) = \sum_{\mathbf{x}} P(\mathbf{x}) \log \left(\frac{p(\mathbf{x})}{R_\lambda(\mathbf{x})} \right)$$

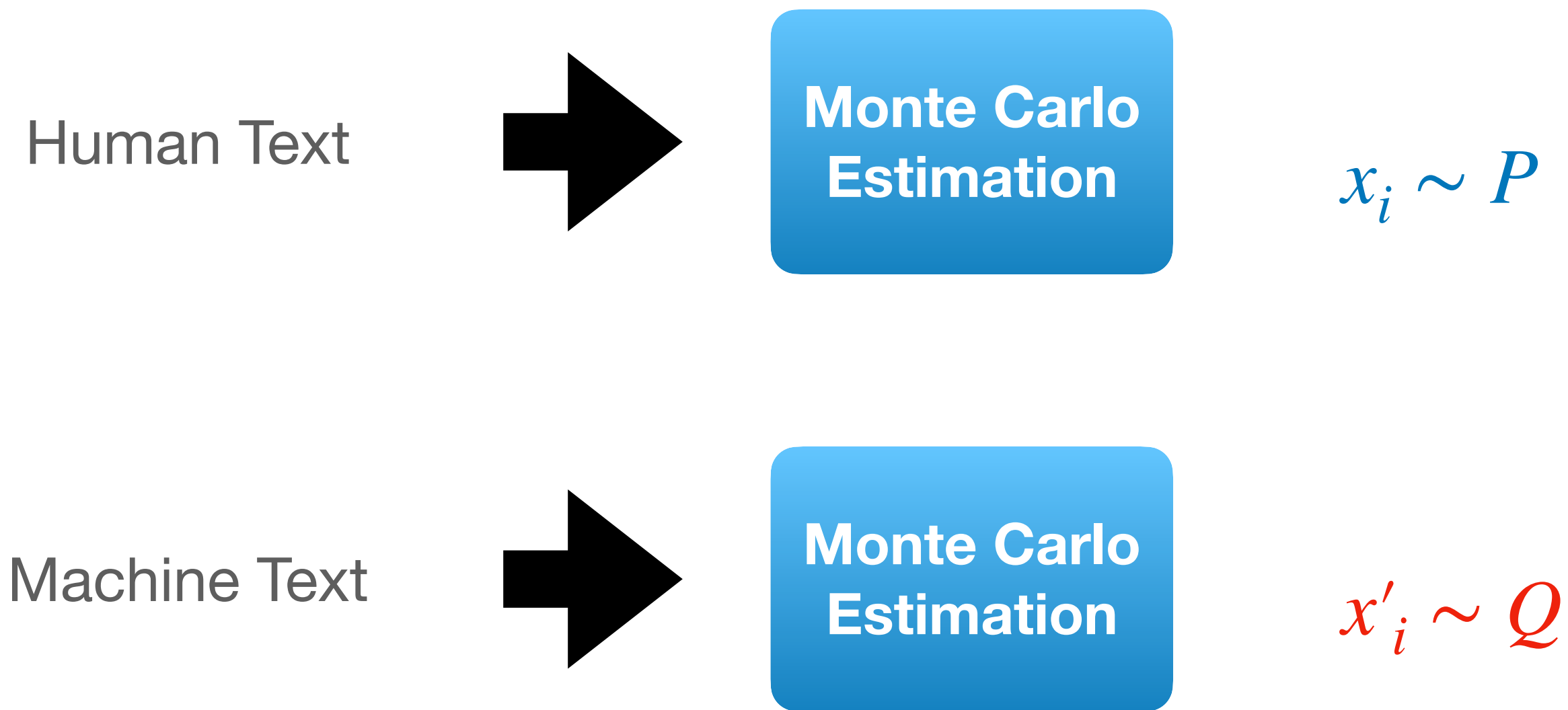
We do not know ground truth probability **P**

Support is prohibitively large, space of all sequence of tokens

Computing **MAUVE** in practice

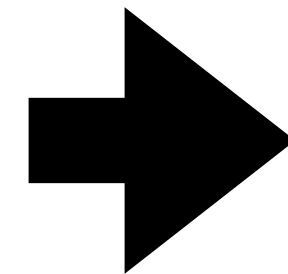


Estimation of ground truth probability

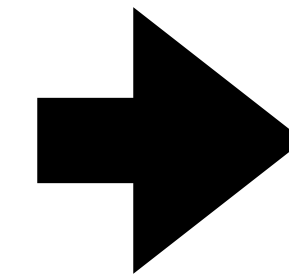


Embedding space

Human
Text Sequence



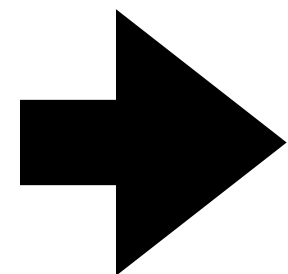
Large
Language
Model



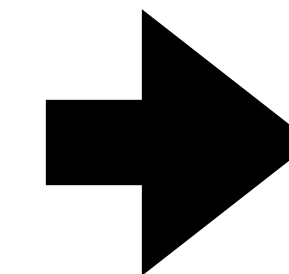
Embedding $M(x)$

d- dimension embedding

Machine
Text Sequence

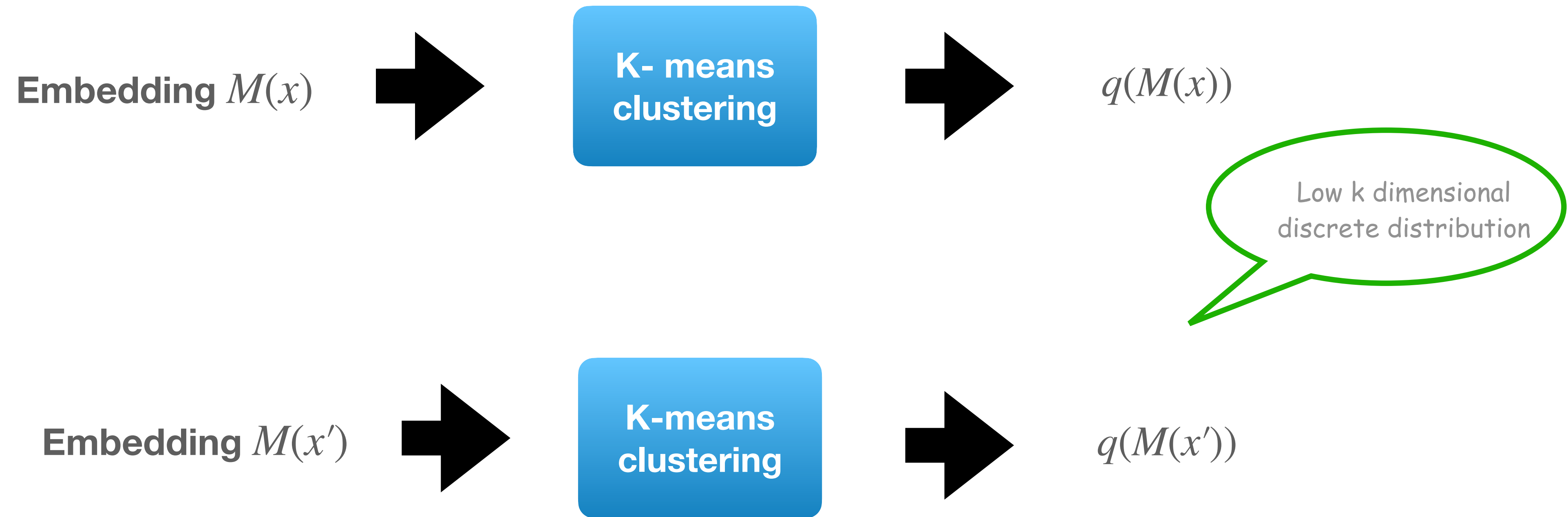


Large
Language
Model



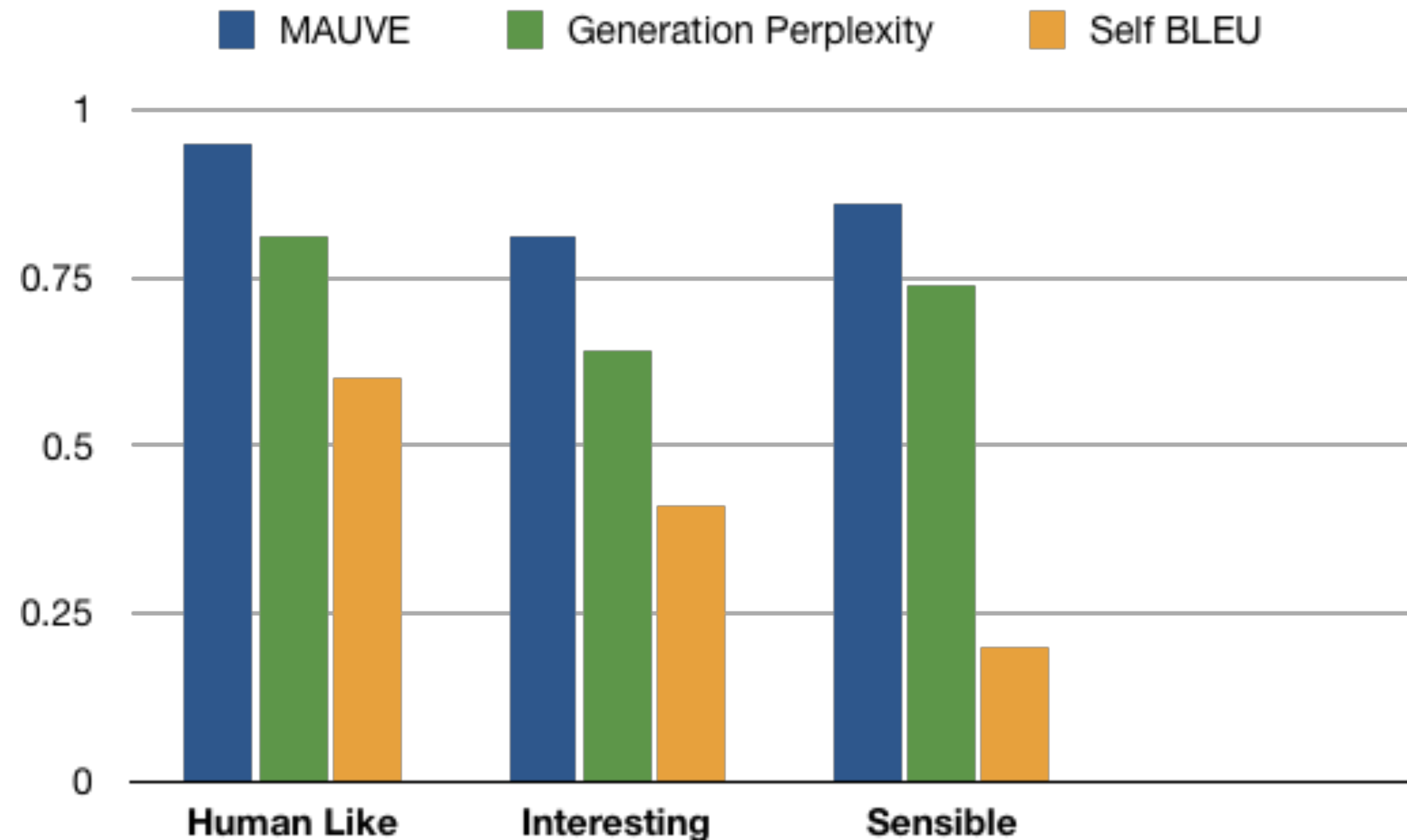
Embedding $M(x')$

Vector Quantization



Mauve correlates with human judgements

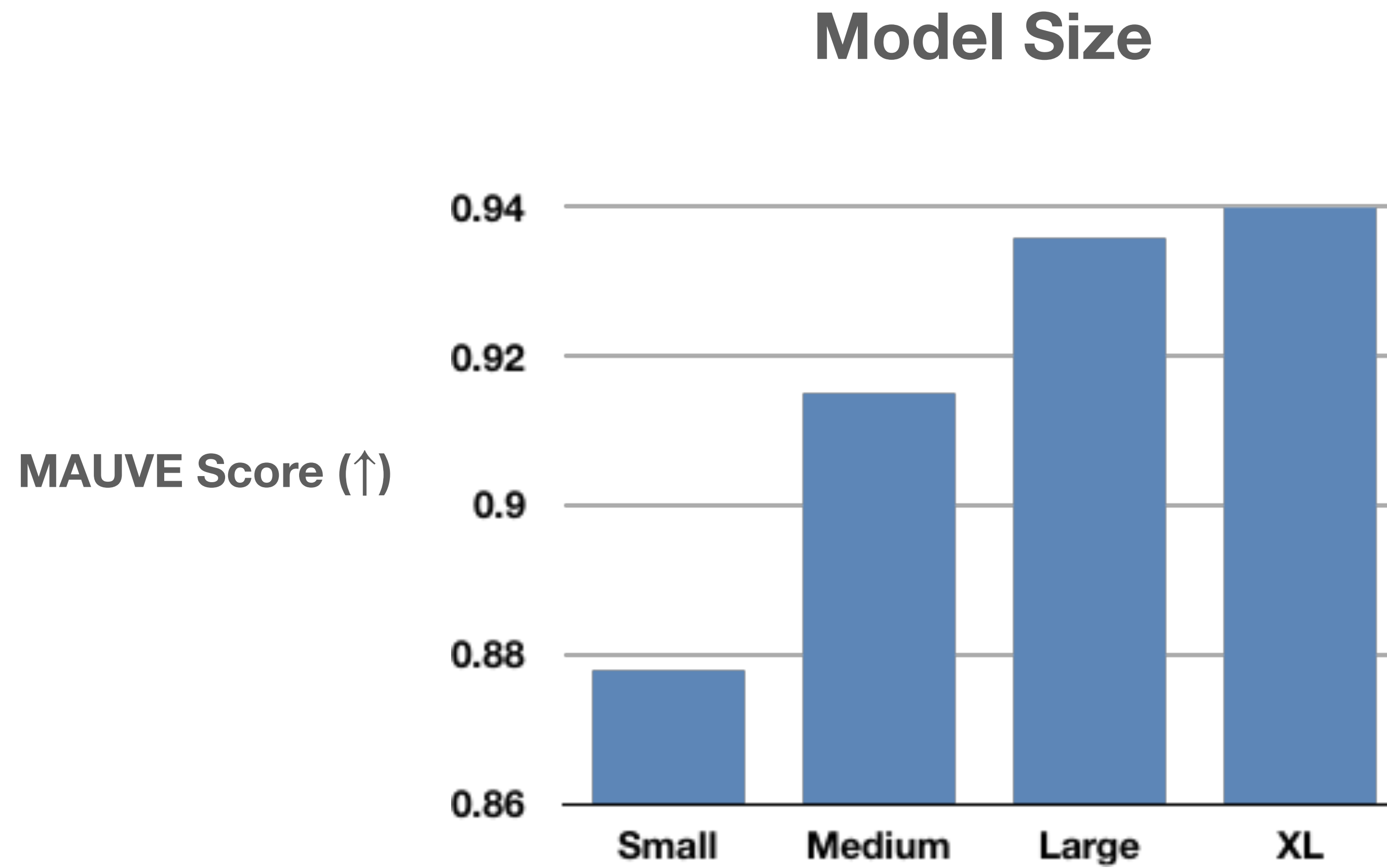
Spearman Correlation with human eval (↑)



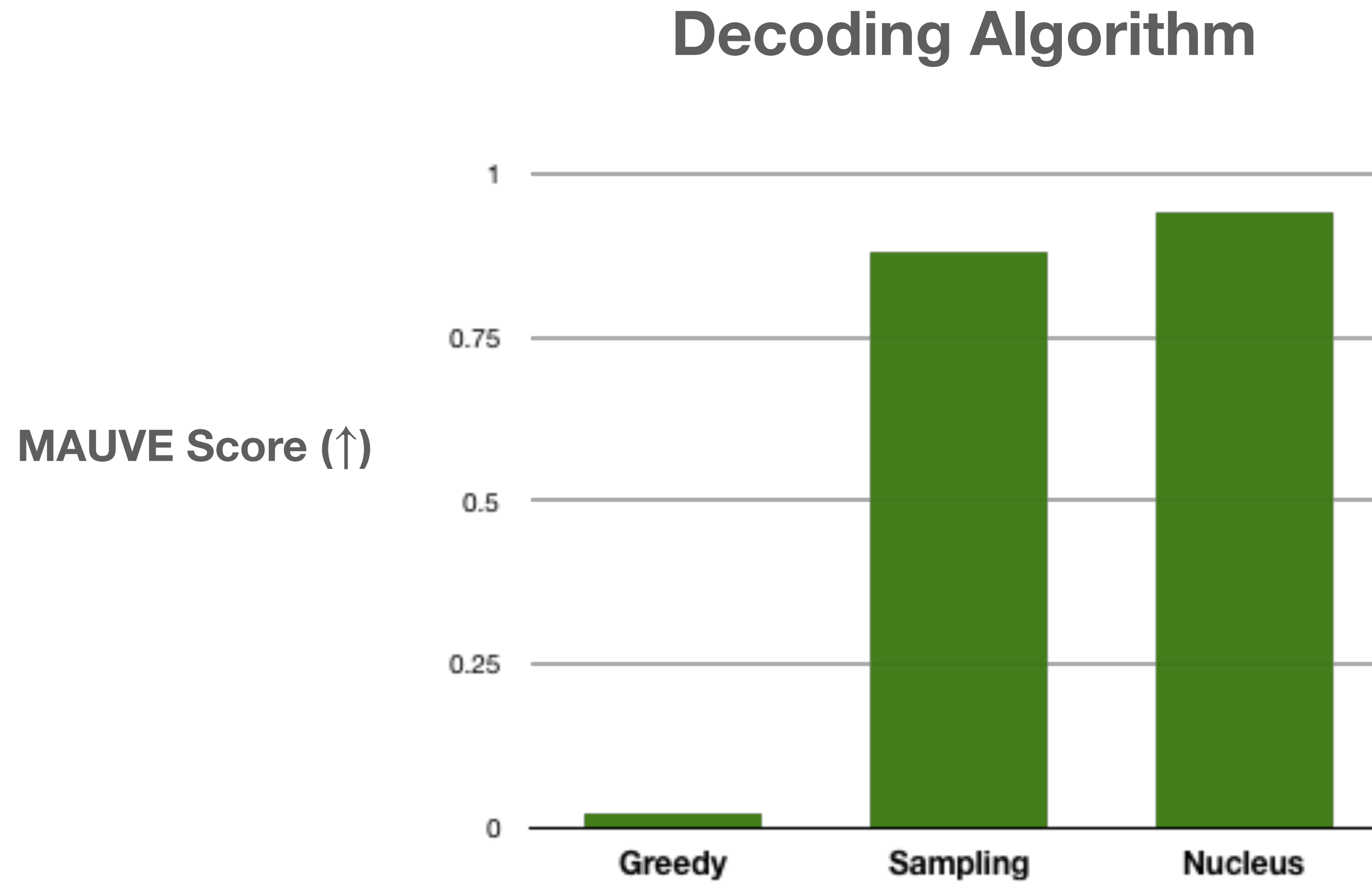
Gen. PPL : Holtzman et al. (ICLR 2020).

Self-BLEU: Zhu et al. (2018)

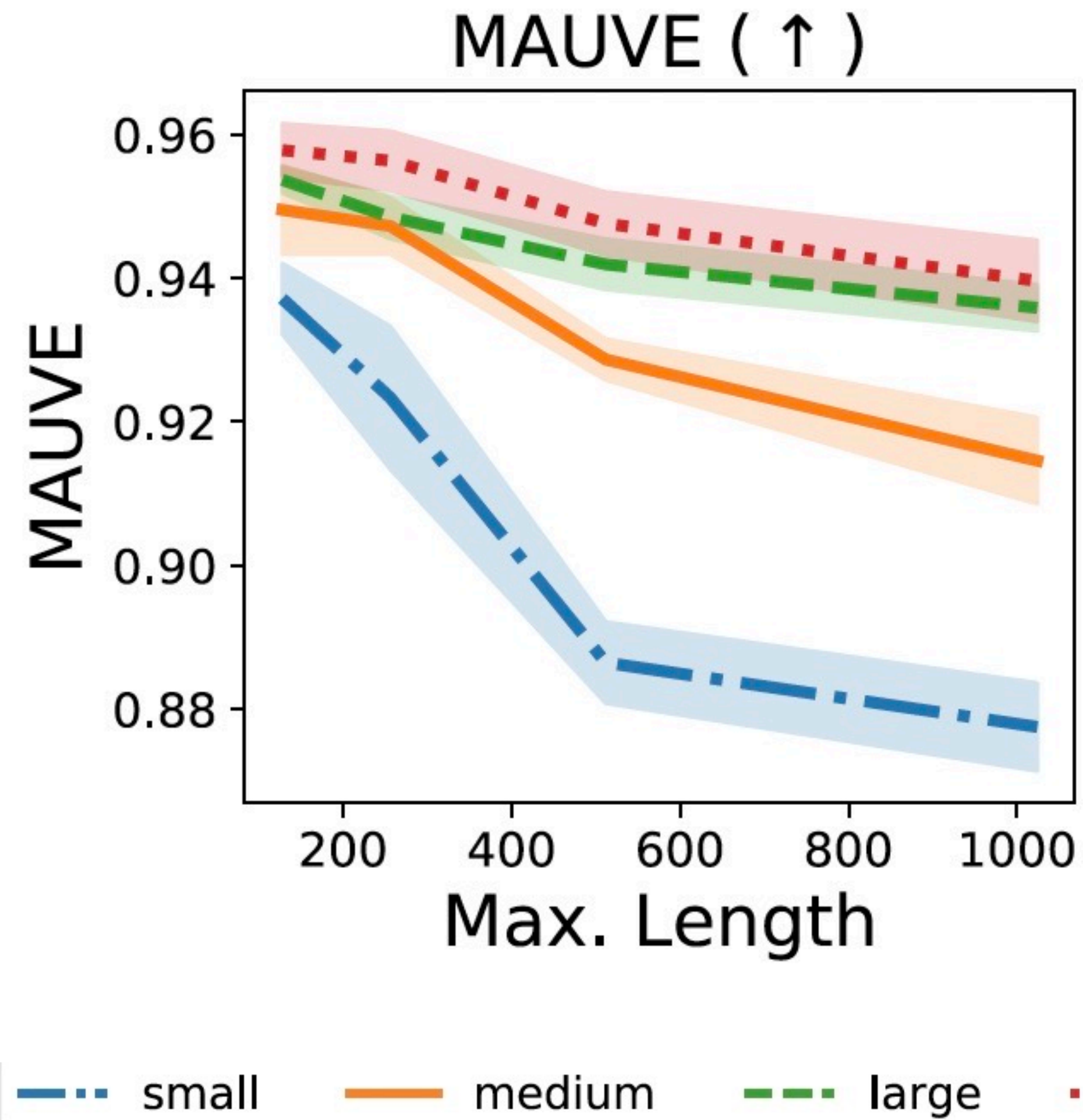
Mauve captures important trends



Mauve captures important trends



Mauve captures important trends



[On to The next Paper]

Published as a conference paper at ICLR 2023

ON THE USEFULNESS OF EMBEDDINGS, CLUSTERS AND STRINGS FOR TEXT GENERATOR EVALUATION

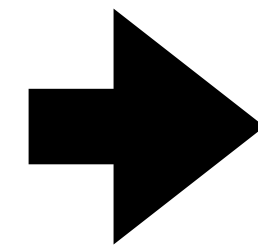
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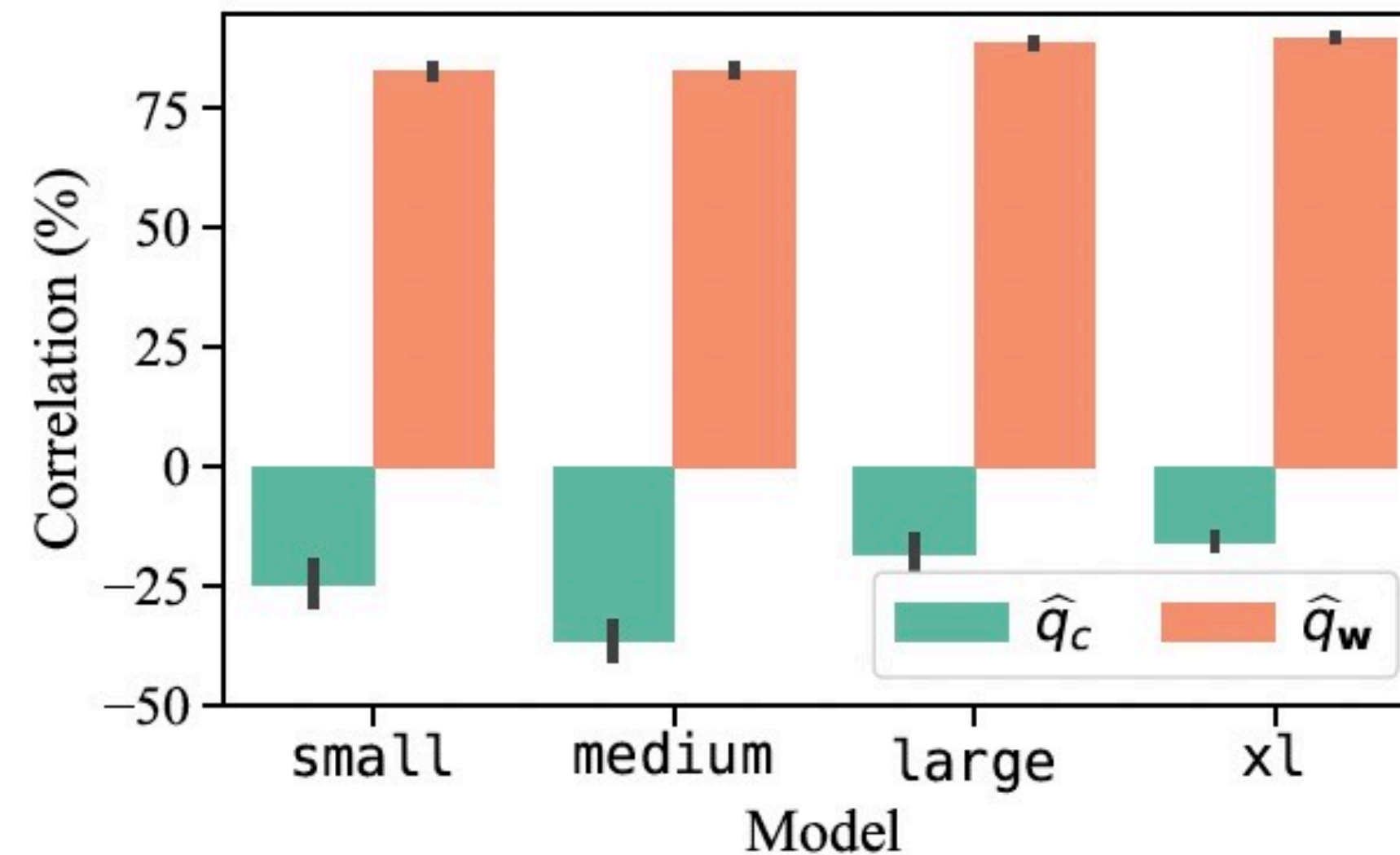
Does P_c approximate P_w ?

Human



Smooth 5 gram
(KenLM)

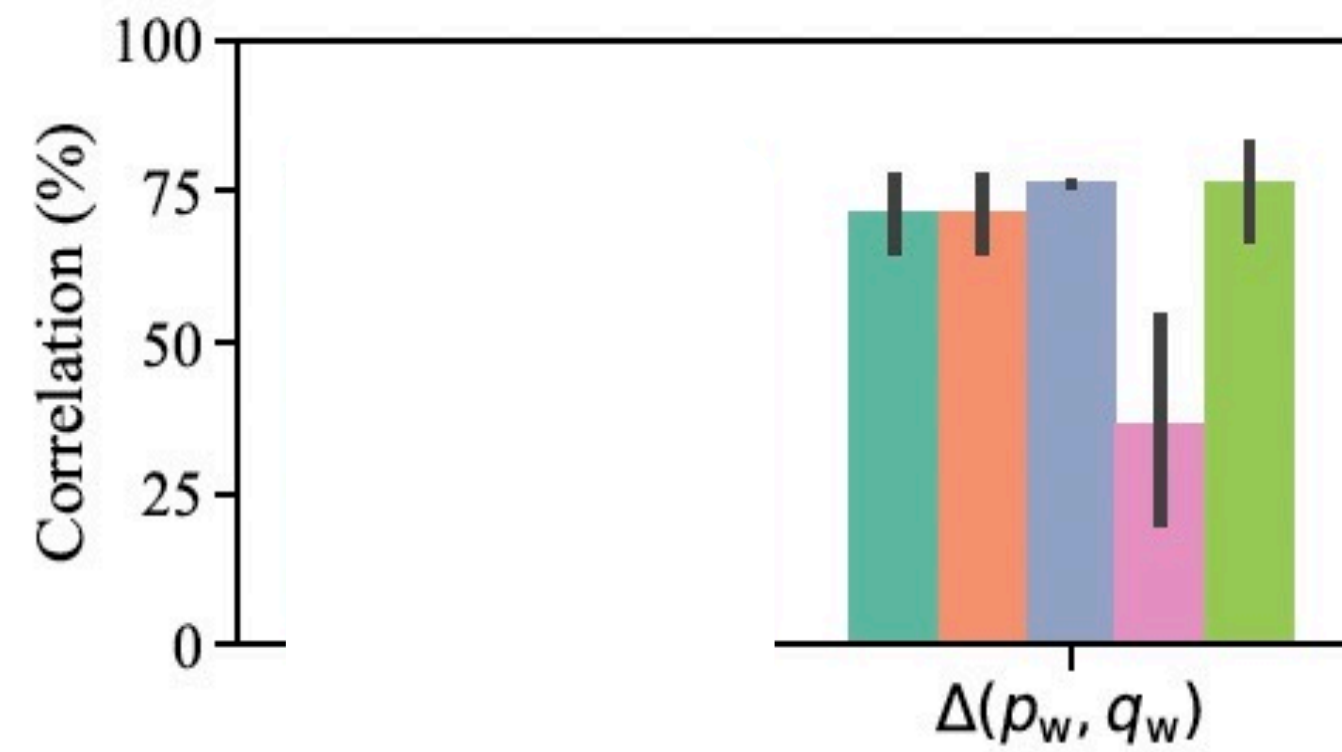
$$x_i \sim P$$



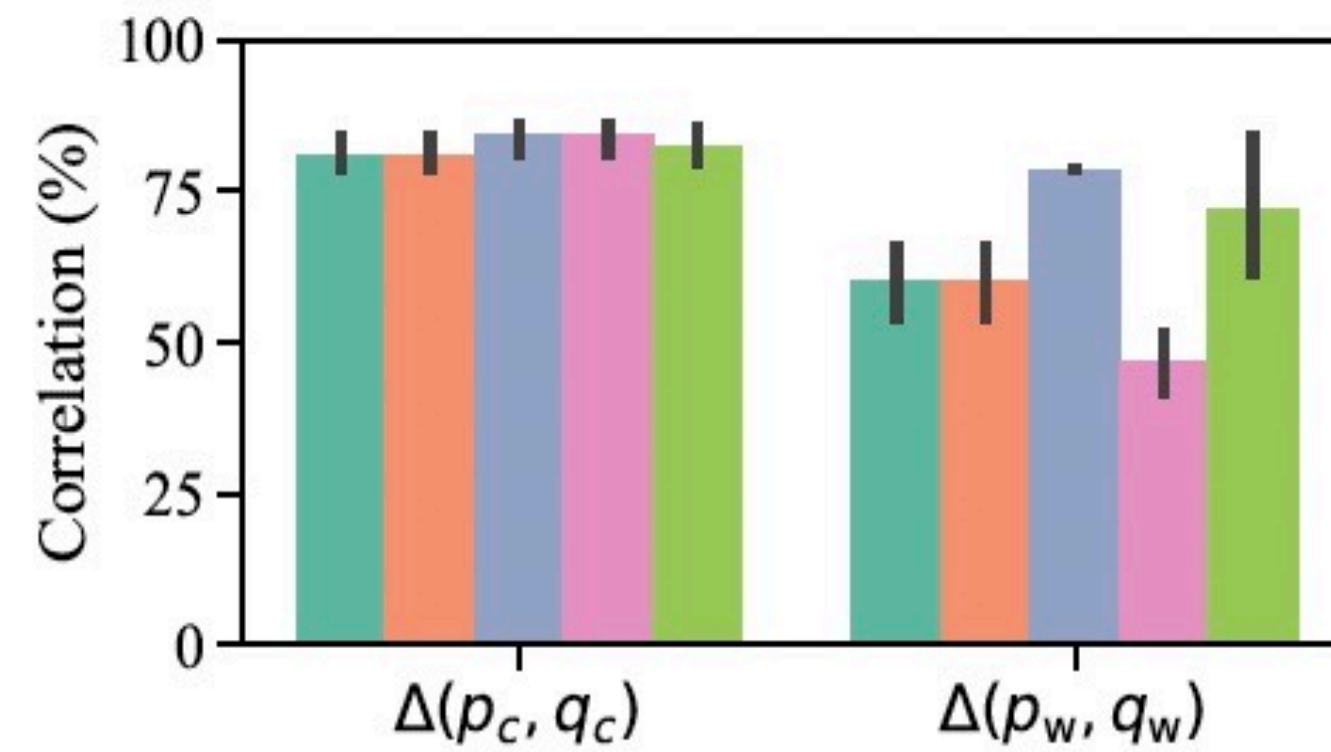
Correlations between true q_w and the estimated $\hat{q}_c$ and $\hat{q}_w$

Is their proposed new divergence really responsible for MAUVE's Success?

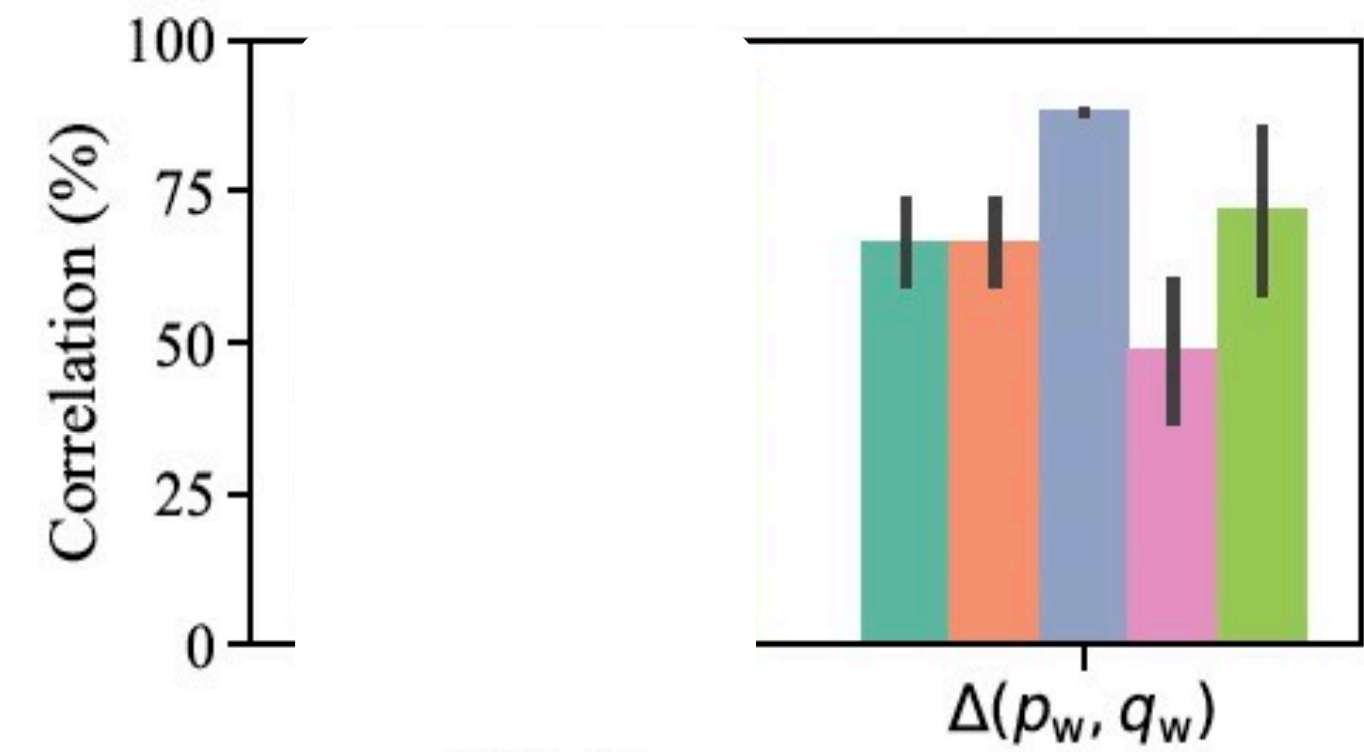
Divergence's usefulness



(a) Interestingness



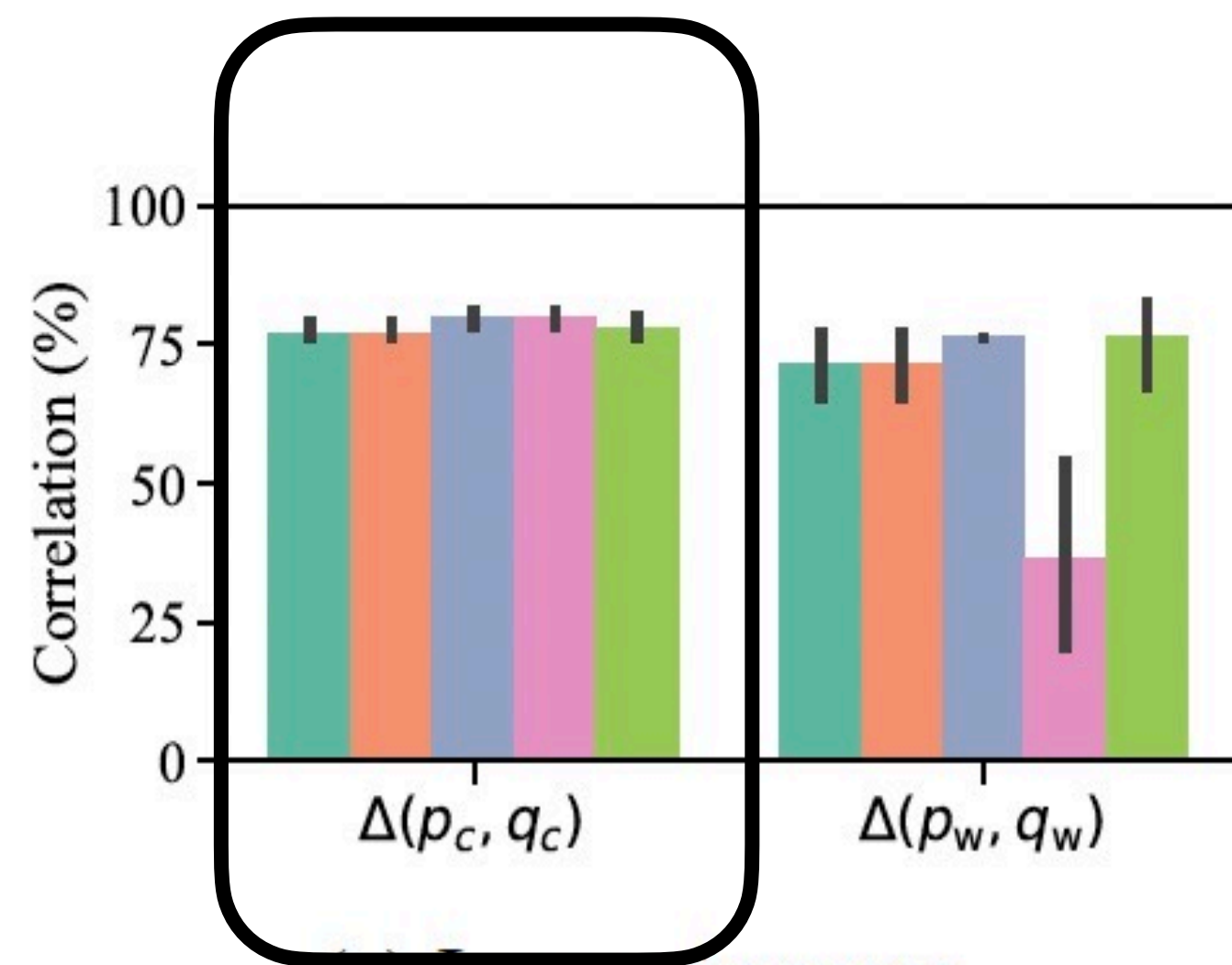
(b) Sensibility



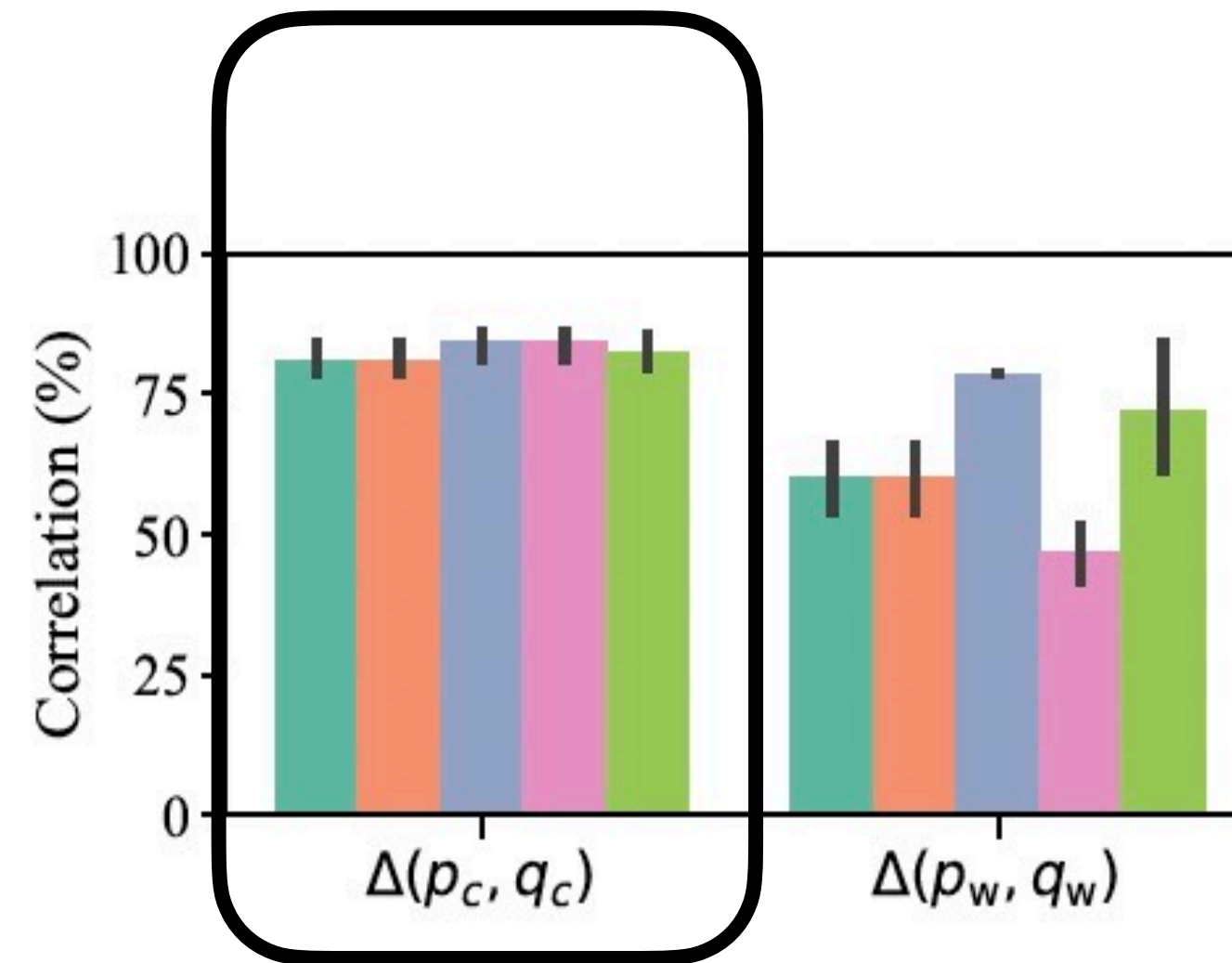
(c) Human likeness

Δ_{exp} in dark green; $\Delta_{\rightarrow}$ in orange; $\Delta_{\leftarrow}$ in blue; Δ_{JS} in pink; Δ_{AUC} in lime green.

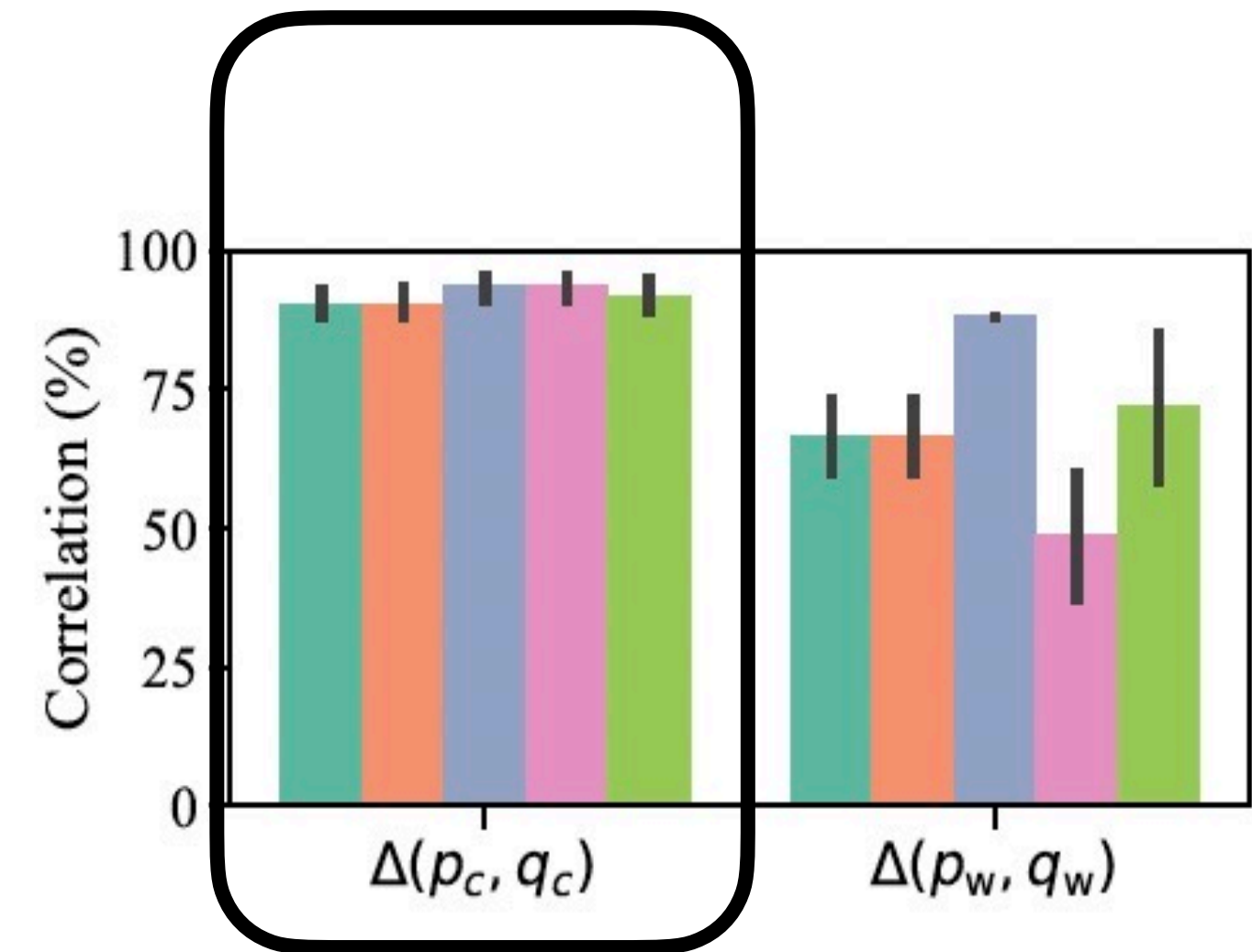
Divergence's usefulness



(a) Interestingness



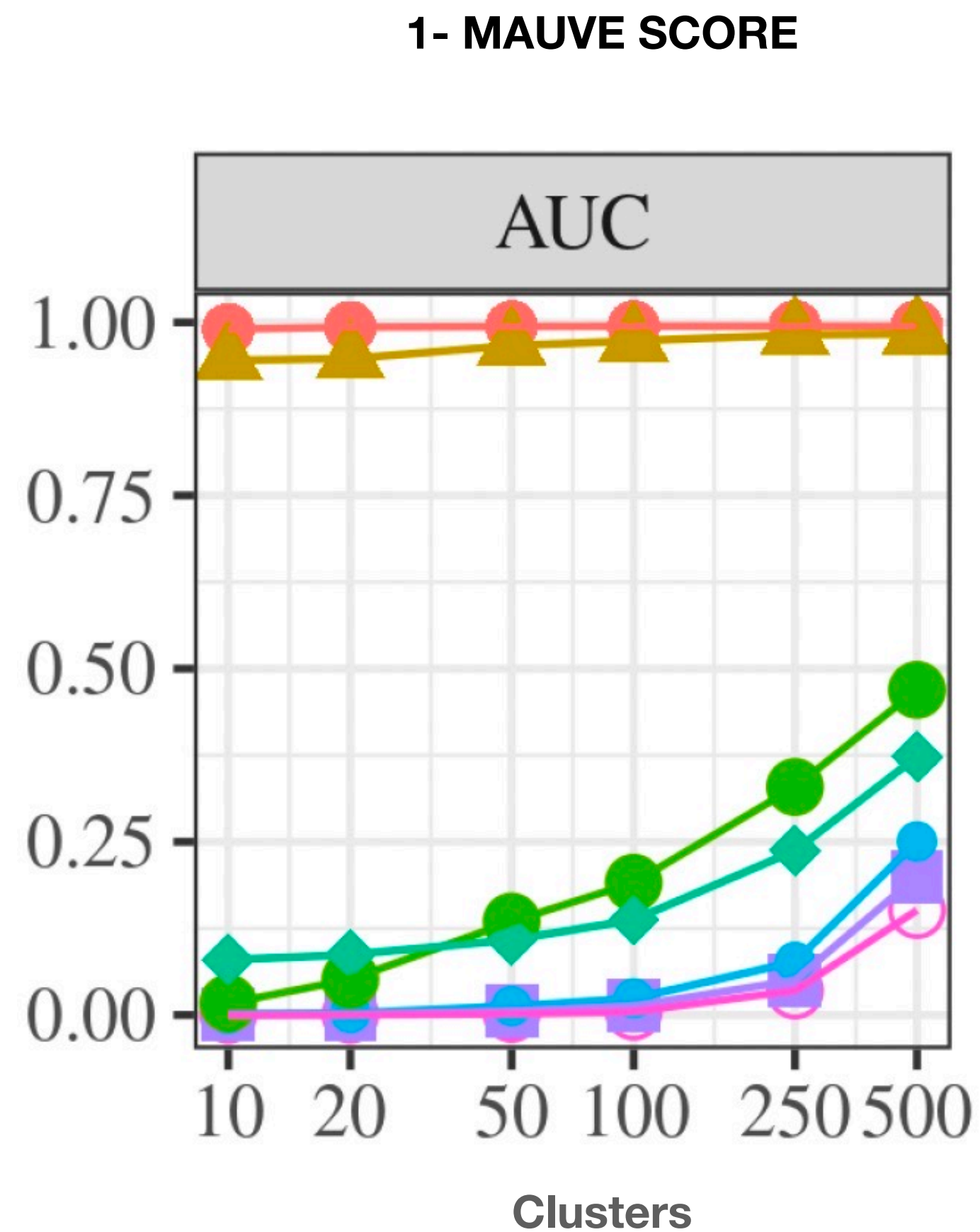
(b) Sensibility



(c) Human likeness

Δ_{exp} in dark green; $\Delta_{\rightarrow}$ in orange; $\Delta_{\leftarrow}$ in blue; Δ_{JS} in pink; Δ_{AUC} in lime green.

What breaks Cluster based evaluators



Comparison Text

p_{rand}

$p_{\text{no stop}}$

p_{swap}

p_{short}

q

$p_{\text{no art}}$

p

Word level permute

Stopwords removal (so, that...)

Sentence level permutation

Text truncation

Article removal

No modification

Cares about coherence & symantics

Don't care about surface statistics